

On the Complexity of Interactive Arguments

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Abstract

We study the minimal hardness assumptions required for constructing *interactive arguments* for NP, focusing on *succinct* arguments—where the prover’s total communication is smaller than the witness size—and on *relaxed* forms of *zero knowledge*, such as *witness indistinguishability*. Known constructions of succinct arguments rely on various types of *collision-resistant hash functions*, *indistinguishability obfuscation*, *hardness of discrete logarithm*, and *lattice-based assumptions*, while known constructions of witness-indistinguishable arguments require *one-way functions* (OWFs). This suggests that succinct witness-indistinguishable interactive arguments require OWFs, or equivalently, that the existence of such arguments implies that of OWFs.

Nevertheless, we prove that, at least as far as fully black-box reductions are concerned, interactive arguments do not imply OWFs. Specifically, we consider *assumption-dependent* fully black-box reductions from OWFs to succinct witness-indistinguishable interactive arguments *and* an additional hardness assumption G (e.g., $\mathsf{NP} \not\subseteq \mathsf{P/poly}$). Such a reduction is a pair (f, R) of (oracle-aided) function f and algorithm R such that, for any witness-indistinguishable interactive argument $\Pi = (\mathsf{P}, \mathsf{V})$ and any inverter Inv of f^Π , the algorithm $\mathsf{R}^{\Pi, \text{Inv}}$ breaks *either* the soundness of Π *or* the assumption G . We prove that the existence of such a reduction implies a black-box reduction from OWFs to G *alone*. Namely, beyond what is implied by G , succinct witness-indistinguishable interactive arguments have no black-box implications for the existence of OWFs.

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1 Introduction

In an *interactive argument* for NP, the prover, holding the witness, interacts with the verifier to convince it of the validity of a statement given as a common input. We require the protocol to be *sound*: even a malicious *efficient* prover cannot make the verifier accept a false statement, except with negligible probability.¹

In *succinct* interactive arguments for NP, the prover gets a witness w as a private input, and the total number of bits it sends is significantly smaller than $|w|$, to the extreme of being poly-logarithmic in $|w|$. Succinct interactive arguments are cornerstones of modern cryptography, enabling applications such as verifiable computation, cf., [GKR08], where a computationally bounded client delegates work to a powerful server while maintaining assurance of the result’s validity. While non-succinct (non-interactive) arguments trivially exist for any NP language—the prover simply sends the witness it receives as private input to the verifier—this does not seem to be the case for succinct arguments; known constructions of succinct interactive arguments for NP are based on *collision-resistant hash functions* [Kil92], *multi-collision resistant hash functions* [KNY18], *indistinguishability obfuscation* [SW14], *hardness of discrete logarithm* [Boo+16], and lattice-based assumptions [Bau+18]. This state of affairs raises the question of whether succinct interactive arguments for NP can be based on *weaker* assumptions, for example, the existence of one-way functions? And the following complementary question, which is in the focus of this paper,

Are one-way functions *required* for building *succinct* interactive arguments for NP?

An interactive argument is *witness indistinguishable* if, after the interaction, the verifier cannot tell which witness the prover used. It is *witness hiding* if, after the interaction, the verifier cannot compute any function of the witness used by the prover better than it could without the interaction. While all known constructions of witness indistinguishable and witness hiding arguments use one-way functions, cf., Goldreich et al. [GMW91], unlike in the case of *zero knowledge*, it is not known that one-way functions are necessary. This raises the following question,

Are one-way functions *necessary* for constructing *witness-indistinguishable* and *witness-hiding* interactive arguments for NP?

1.1 Our Results

We study the existence of *fully black-box reductions* from one-way functions (OWFs) to interactive arguments. It is easy to see that, unless one proves that $P \neq NP$, reductions from one-way functions to the existence of succinct interactive arguments do *not* exist; if $P = NP$, then one-way functions do not exist, but non-interactive arguments do exist (the verifier decides the language by itself). Hence, a reduction from one-way functions to the existence of interactive arguments (of any type) would establish that $P \neq NP$. Therefore, the right type of reductions to consider is from one-way functions to interactive arguments *together with* an additional hardness assumption, e.g., $NP \not\subseteq P/\text{poly}$. Such reductions would enable, for instance, to prove that one-way functions are necessary for building succinct or witness-indistinguishable interactive arguments for *hard-*

¹Interactive arguments are a relaxation of *interactive proofs*, where soundness is required to hold against *arbitrary* malicious provers.

to-decide languages.² To formalize such reductions, we naturally capture hardness assumptions using interactive games between a *player* and a (possibly inefficient) *challenger*,³ and consider the following type of *assumption-dependent* reductions.

Definition 1.1 (Assumption-dependent, fully black-box reductions from OWFs to interactive arguments, informal). *An assumption-dependent, fully black-box reduction from OWFs to interactive arguments for an NP-relation $\mathcal{R}$, with respect to a game G , is a poly-time, oracle-aided function f and a PPT, oracle-aided algorithm R , such that for every efficient interactive argument (P, V) and any efficient Inv that inverts $f^{P, V}$ with non-negligible probability, $R^{(P, V), \text{Inv}}$ breaks the soundness of (P, V) , with respect to $\mathcal{R}$, or wins the game G , with non-negligible probability.*

That is, if f is not one-way then either the interactive argument (P, V) is not sound or the game G is not hard. We make the following remarks:

- The one-way function f treats the interactive argument parties as oracles, but might depend *arbitrarily* on the game G .
- The security proof R treats both the interactive argument and an adversary inverting f as oracles.
- We only require the reduction to work given access to an *efficient* interactive argument and an efficient inverter, where “efficient” stands for non-uniform poly-time algorithms.⁴

For such reductions, we prove the following result.

Theorem 1.2 (Informal). *If there exists an assumption-dependent, fully black-box reduction from OWFs to interactive arguments for an NP-relation $\mathcal{R}$, with respect to a game G , then there exists a black-box reduction from OWFs to G . Moreover, the above holds even if the reduction only needs to work when the interactive argument oracle is perfect zero knowledge.*

Namely, if one can use the soundness of an interactive argument (succinct or not), together with the hardness of a game G to construct one-way functions, then one could also use *only* the hardness of G for this task. Moreover, since being perfect zero knowledge implies being perfect witness indistinguishable and witness hiding, the above holds also for reductions from one-way functions to witness-indistinguishable/hiding arguments. Note, however, that Theorem 1.2 does *not* address reductions that get an oracle access to the zero-knowledge simulator.⁵ Theorem 1.2 naturally extends to *auxiliary-input* one-way functions.

²Similar types of reductions were used by Ostrovsky and Wigderson [OW93] and Hirahara and Nanashima [HN24] to prove that the existence of one-way functions is necessary for building *zero-knowledge proofs* for hard-to-decide languages. See more details in Section 1.3.

³Our actual definition, see Section 2.5, allows more complex games than such interactive protocols, and captures a larger set of hardness assumptions, such as $\text{NP} \not\subseteq \text{P/poly}$.

⁴Most separation results for fully black-box reductions exploit the fact that the reduction under consideration is required to work given access to *any* instantiation of the hardness assumption and the adversary. Requiring the reduction to succeed only given access to *efficient* entities simplifies the reduction’s task, and thus makes lower bounds on such reductions, such as the ones we present here, more insightful.

⁵This is mandatory, since there *are* black-box reductions from zero-knowledge arguments and hardness of NP, to one-way functions. See Footnote 2 and section 1.3.

1.2 Our Technique

Fix an assumption-dependent, fully black-box reduction (f, R) with respect to NP-relation $\mathcal{R}$ and a game G . Our approach is to give a construction of an efficient non-interactive argument Π ,⁶ and an efficient (no oracle) function g , such that inverting g implies the inversion of $f^\Pi := f^{P, V}$. Hence, for any inverter Inv of g , it holds that $R^{\Pi, \text{Inv}}$ either breaks the soundness of Π , or wins the game G . By carefully constructing the argument Π , we prove that the former does not hold, and conclude that $R^{\Pi, \text{Inv}}$ wins the game G . Finally, by letting R emulate Π internally (recall that Π is efficient), we get an efficient algorithm $\hat{R}$ such that $\hat{R}^{\text{Inv}}$ wins the game G . Namely, we construct a (black-box) reduction from the one-wayness of g to the hardness of G . (We ignore the zero-knowledge part for now, and address it in Section 1.2.2.)

Given the above blueprint, the main challenge is to construct the desired argument Π . We start with a simple approach, explain its shortcomings, and gradually develop the actual argument. Throughout, we assume without loss of generality that f only makes valid queries $(\phi, w) \in \mathcal{R}$ to the prover (by design, our prover returns $\perp$ on invalid queries). We also assume for simplicity that both f and R are *length-preserving*: on inputs of length n , the function f queries the interactive argument Π only on instances of length n , and the reduction R queries Π and the potential adversary Inv only on inputs of this length. In Section 1.2.1, we explain how to handle arbitrary reductions.

The unpredictable-function argument. When considering interactive arguments for an NP-relation $\mathcal{R}$ to be given as oracles to the reduction, the first thing that might come to mind is to use an *unpredictable function*. That is, consider the following argument $\Pi_h = (P_h, V_h)$.

Prover P_h . On input (ϕ, w) : If $(\phi, w) \notin \mathcal{R}$, return $\perp$. Return $h(\phi)$.

Verifier V_h . On input (ϕ, π) : Accept if and only if $\pi = h(\phi)$.

Here, $h: \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ is sampled from a family of $t(n)$ -wise independent hash functions, where $t(n) \in \text{poly}(n)$ is larger than the query complexity of f and R , and $\ell(n) \in \omega(\log n)$. While the above argument uses shared randomness (i.e., h), it can later be replaced with fixed non-uniform advice. Clearly, the above argument is perfectly complete (for any fixing of h). The problem is that we do not know how to prove that an inverter Inv of f^{Π_h} , cannot be used by the reduction R to break the soundness of Π_h ,⁷ and our proof strategy does not go through.⁸

⁶Using non-interactive arguments in our proof stipulates that the black-box reduction cannot benefit from the low round complexity of the argument.

⁷While the query complexity of f and R is smaller than t , the information returned by Inv might be used by R to break the unpredictability of h , thus breaking the soundness of Π_h .

⁸Interestingly, if we use a truly random function for h , i.e., a random oracle, then one can prove that $R^{\Pi_h, \text{Inv}}$ cannot break the soundness of Π_h . It is unclear, however, how to define the efficient one-way function g with respect to such an inefficient argument.

Separately answering f -generated queries. The crux of our approach is modifying the above argument so that it no longer uses h on “inputs generated by f ”. By doing so, we are guaranteed that an inverter of f^{Π_h} will not be useful in breaking the soundness of Π_h , overcoming the shortcoming of the first construction of Π_h . As a first simple case, assume that the function f makes only a single query (ϕ, w) to the prover P_h (and none to the verifier), and let $\mathbf{qry}$ denote the function that maps $x \in \{0, 1\}^n$ to the first part of the query (i.e., the statement part). A key observation is that we can assume without loss of generality that $\mathbf{qry}$ is *not* one-way; if it is one-way, let $g := \mathbf{qry}$ and let $\hat{R}$ be the reduction that does nothing. Since g is one-way, there is no efficient Inv that inverts it. Thus, $\hat{R}$ is a valid reduction to the security of G , and the pair $(g, \hat{R})$ concludes the proof. For the same reason, we can assume that $\mathbf{qry}$ is not even weakly one-way.⁹ This leads to the following interactive argument. Let $\mathbf{stInv}$ be a *strong inverter* of $\mathbf{qry}$: it finds a preimage of $\mathbf{qry}$ with all but negligible probability (since, by assumption, $\mathbf{qry}$ is not weakly one-way, such an $\mathbf{stInv}$ exists).

Prover P_h . On input (ϕ, w) : If $(\phi, w) \notin \mathcal{R}$, return $\perp$.

1. Let $z \leftarrow \mathbf{stInv}(\phi)$.
2. If $\mathbf{qry}(z) = \phi$, return “check”. Otherwise, return $h(\phi)$.

Verifier V_h . On input (ϕ, π) : If $\pi \neq \text{“check”}$, accept if and only if $\pi = h(\phi)$.

Otherwise:

1. Let $z \leftarrow \mathbf{stInv}(\phi)$.
2. Accept if and only if $\mathbf{qry}(z) = \phi$.

That is, the prover lets the verifier *decide by itself* on instances generated by f . The key observation is that when called by f , it will almost always be the case that the prover identifies the query as such, and will return “check”. Hence, the function f (effectively) *gets no access to h* . Let g be the efficient function that on input x , emulates $f(x)$ while answering all P_h queries with “check”, and let Inv be an inverter of g . Namely, g captures the behavior of f , but under simulated oracle answers that avoid access to h . Since the output distributions of g and f^{Π_h} are statistically close, Inv also inverts f^{Π_h} . The point is that being (effectively) *independent of h* , algorithm R^{Inv, Π_h} cannot break the soundness of Π_h , and thus it must win G . Hence, the function g and the reduction $\hat{R}$, which acts like R while emulating the calls to Π_h internally, establish the proof.

Multiple prover-queries. The above naturally extends to the case where f makes *multiple* queries to P (but no queries to V). Define the function $\mathbf{qry}(x, i)$ to output the i^{th} query made by $f^{P_h}(x)$, while answering its previous P_h -queries with “check”. Define the function g analogously to $\mathbf{qry}$ while outputting f ’s (final) output (and not its i^{th} query). Finally, let $\mathbf{stInv}$ be a strong inverter of $\mathbf{qry}$. By construction, the output distributions of g and f^{Π_h} are close, and a similar proof to the one in the single-query case goes through.

⁹Weak one-way functions can be amplified, in a fully black-box manner, to full-fledged one-way functions, and the same argument as above would apply.

Verifier queries. We finally explain how to handle queries made by f to the verifier V_h . For the above construction, allowing verifier queries would give the function f access to an inverter of its query function $\mathbf{qry}$. This would yield a recursive definition of $\mathbf{qry}$, rendering the argument inefficient for a function f of super-constant query complexity. We handle this problem by using a second hash function h^C for the “easy” (i.e., f -generated) queries, preventing f from making verifier queries for which it does not already know the answer.

We define the argument Π using a query function $\mathbf{qry}$ that on input (x, i) outputs the statement part of the i^{th} query made by $f(x)$, while answering f ’s oracle queries as follows:

P-query $(\phi, \cdot)$. Answer $h^C(\phi)$.

V-query (ϕ, π) . Answer 1 if and only if $\pi = h^C(\phi)$.

Let $\mathbf{stInv}$ be an efficient, strong inverter of $\mathbf{qry}$. (Note that $\mathbf{qry}$ is clearly efficient.)

Prover P_{h^C, h^T} . On input (ϕ, w) : If $(\phi, w) \notin \mathcal{R}$, return $\perp$.

1. Let $z \leftarrow \mathbf{stInv}(\phi)$.
2. If $\mathbf{qry}(z) = \phi$, return $h^C(\phi)$. Otherwise, return $h^T(\phi)$.

Verifier V_{h^C, h^T} . On input (ϕ, π) :

1. If $\pi = h^T(\phi)$, accept. Otherwise,
2. Let $z \leftarrow \mathbf{stInv}(\phi)$.
3. Accept if and only if $\pi = h^C(\phi)$ and $\mathbf{qry}(z) = \phi$.

That is, on f -generated queries, the prover returns $h^C(\phi)$, and the verifier checks that $\pi = h^C(\phi)$. Define the function g analogously to $\mathbf{qry}$ while outputting f ’s (final) output. By construction, the output distributions of g and $f^{\Pi_{h^C, h^T}}$ are close, and a similar proof to the one in the prover-only queries case goes through.

1.2.1 Handling Arbitrary Reductions

When considering arbitrary reductions, the function f can query Π (i.e., the prover and verifier) on queries of arbitrary length. Hence, we no longer can instruct the query function $\mathbf{qry}$ to return $h^C(\phi)$ on all prover-queries of the form $(\phi, \cdot)$; if the query (ϕ, w) has length $n' \leftarrow \log n$ (where n is the input length of f), then in terms of n' , the function f has invested *exponential time* in generating it, and the prover will use the function h^T on such queries. So to closely mimic the real query function of f , the query function $\mathbf{qry}$ has to also use h^T , and our proof strategy breaks down.

The solution is to change Π to use a different pair of hash functions (h^C, h^T) according to the length of ϕ . In addition, we instruct P and V to try, on input $(\phi, \cdot)$, to invert the query function, which now is using h^T , on *all* inputs of length $n^{1/c}, \dots, n^c$, for an appropriate choice of $c \in \mathbb{N}$, and for $n \leftarrow |\phi|$. Proving the above turned out to be rather subtle, and constitutes the most technically challenging part of the paper.

1.2.2 Zero Knowledge

Note that the interactive arguments we considered above are *perfect* zero-knowledge. Indeed, on input ϕ , the simulator simply acts as the prover P would on input (ϕ, w) , after verifying that $(\phi, w) \in \mathcal{R}$. Consequently, our implication—that any fully black-box reduction from one-way functions to interactive arguments (possibly combined with G) yields a reduction from one-way functions to G alone—continues to hold even if the reduction (f, R) is only required to work when the interactive arguments are *perfect* zero-knowledge.

As a sanity check, observe that if R was given access to the simulator of Π , it could use this access to break the soundness of Π . However, our definition of fully black-box reductions does not permit such access.

1.3 Related Work

We briefly survey work related to our result.

Black-box limitations. The use of black-box limitations in cryptography was initiated in the seminal work of [IR89], showing a *black-box separation* between key-agreement protocols and one-way functions.¹⁰ Subsequent work established black-box limitations on constructions of other primitives, such as collision-resistant hash functions [Sim98] and oblivious transfer [Ger+00]. Our proof technique resembles the so-called *meta-reductions*, cf., [BMV08; HRS09], where one shows that the oracle gates used by the black-box proof of security (i.e., R in our case) can be *emulated* efficiently.

Constructing one-way functions. One-way functions are known to be implied by the existence of many other primitives and hardness assumptions, e.g., pseudorandom generators, cf., [IL89]. Among them, the most relevant results are the construction of one-way functions from the existence of *zero-knowledge* (ZK) proofs/arguments for (all) NP together with the assumption that NP is hard. [OW93] pioneered this direction by constructing auxiliary-input one-way functions assuming ZK-proofs (or arguments) for NP and $NP \not\subseteq BPP$. More recently, [HN24] (see also [LMP24]), improved this result by showing the existence of (standard) one-way functions, assuming ZK-arguments for NP and $NP \not\subseteq i.o.P/poly$.¹¹ The above constructions are *assumption-dependent fully black-box reductions from one-way functions to ZK proofs/arguments, the zero-knowledge simulator, and the relevant hardness assumption* (under the natural adaptation of Definition 1.1 to zero-knowledge protocols).

Succinct arguments. The feasibility of succinct interactive arguments was established by [Kil92], who constructed such arguments for all NP assuming the existence of collision-resistant hash functions. More recently, [KNY18] showed a construction from multi-collision resistant hash functions. Succinct (zero-knowledge) arguments were also constructed using stronger assumptions, such as *discrete logarithm* [Boo+16] and lattice-based assumptions [Bau+18] (see also [Boo+20; ACK21;

¹⁰A black-box separation presents a “world” in which the assumption, e.g., OWFs, exist but the desired primitive, e.g., *key agreement*, does not. Black-box separations are more informative than fully black-box reductions; they can be used, for instance, to refute the existence of a two-part black-box construction in which the second part uses an adversary of the first part, if such exists (see [RV24] for an example of such a construction), where fully black-box reduction cannot.

¹¹The class *i.o.P/poly* contains all languages that can be solved by poly-size circuits, for infinitely many n ’s.

[BCS21; BS23; BCS23]). For succinct *non*-interactive arguments (SNARGs), [GW11] proved that (certain types of) SNARGs cannot be black-box reduced to a falsifiable assumption (as defined by [Nao03]). Following this barrier, SNARGs were constructed using non-falsifiable assumptions, such as indistinguishability obfuscation (io) [SW14] and the generic group model [Gro16] (see also [Mic00; BS+18; Set20; BFS20]). The barrier was also bypassed by considering more restricted languages, specifically by constructing SNARGs for P from falsifiable assumptions such as *bilinear groups* [KPY19] and *learning with errors (LWE)* [CJJ21]. On a different vein, [HNY22] proved that, assuming the randomized exponential time hypothesis (rETH) holds, certain types of SNARGs constructions in the *random-oracle model (ROM)* cannot be “very concise”.

Succinct proofs. Succinctness was also studied in the context of interactive proofs (i.e., soundness should hold against unbounded provers). [GVW01] analyzed the complexity of interactive proofs with succinct (referred there as *laconic*) provers. They showed that any language admitting an interactive proof where the prover sends at most $c(n)$ bits is decidable in probabilistic time $O(2^{c(n)})$. Thus, (sufficiently) succinct interactive proofs are unlikely to exist for all NP.

Witness indistinguishability and witness hiding. [GMW86] were the first to construct zero-knowledge (ZK) arguments (which are also witness indistinguishable and witness hiding) for NP, assuming the existence of one-way functions. [FLS99] showed how to construct non-interactive ZK arguments using *trapdoor permutations*. [DN00] relaxed the required assumption for non-interactive witness indistinguishable arguments (NIWIs) to *verifiable PRGs*. Later, [GOS06] constructed NIWIs from *bi-linear groups*, and [BP15] have done it assuming *indistinguishability obfuscation*.

Paper Organization

Section 2 contains definitions and background material used in the paper. The formal statement of our result and its proof, are given in Section 3.

2 Preliminaries

2.1 Notation

We use calligraphic letters to denote sets, uppercase for random variables, lowercase for integers and functions, and boldface for vectors. All logarithms are base 2. For $n \in \mathbb{N}$, let $[n] := \{1, \dots, n\}$. For a vector $\mathbf{x} := (x_1, \dots, x_n)$ and $j \leq n$, let $(\mathbf{x})_j := x_j$ and let $(\mathbf{x})_{\leq j} := (x_1, \dots, x_j)$. We naturally extend this notation to strings as well. For a distribution $\mathcal{X}$, let $x \stackrel{\mathcal{R}}{\leftarrow} \mathcal{X}$ denote the process of sampling x according to $\mathcal{X}$. Similarly, for a set $\mathcal{S}$, $s \stackrel{\mathcal{R}}{\leftarrow} \mathcal{S}$ stands for the process of sampling s uniformly from $\mathcal{S}$. For two distributions $\mathcal{X}$ and $\mathcal{Y}$, let $\text{SD}(\mathcal{X}, \mathcal{Y})$ denote their *statistical distance* (also known as, variation distance). Let poly stand for the set of all (integer) polynomials. We sometimes abuse notation and treat poly as an arbitrary polynomial. A function $f(n)$ is *negligible* if for every $p \in \text{poly}$ there exists $n_0 \in \mathbb{N}$ such that $f(n) < 1/p(n)$ for all $n > n_0$. We write $f(n) \leq \text{neg}(n)$ to denote that $f(n)$ is negligible. For an NP-relation $\mathcal{R}$, let $\mathcal{L}_{\mathcal{R}}$ be the language $\{x : \exists w \text{ s.t. } (x, w) \in \mathcal{R}\}$. We use PPT to denote *probabilistic polynomial time* and PPTM to denote PPT machines, i.e., algorithms. A circuit family $\mathcal{C} = \{\mathcal{C}_n\}_{n \in \mathbb{N}}$ is of size p if $|\mathcal{C}_n| \leq p(n)$ for every $n \in \mathbb{N}$.

2.2 Many-wise Independent Hashing

Definition 2.1 (Many-wise Independent Hash). A function-family $\mathcal{H} = \{h: \mathcal{X} \rightarrow \mathcal{Y}\}$ is t -wise independent, if for any distinct inputs $x_1, \dots, x_t \in \mathcal{X}$ and any values $y_1, \dots, y_t \in \mathcal{Y}$:

$$\Pr_{h \leftarrow \mathcal{H}} [h(x_1) = y_1 \wedge \dots \wedge h(x_t) = y_t] = 1/|\mathcal{Y}|^t.$$

Definition 2.2 (Efficient function ensemble). A function-family ensemble $\mathcal{H} = \{\mathcal{H}_n = \{h: \mathcal{X}_n \rightarrow \mathcal{Y}_n\}\}_{n \in \mathbb{N}}$ is efficient, if sampling $h \leftarrow \mathcal{H}_n$ and computing $h(x)$ for $h \in \mathcal{H}_n$ and $x \in \mathcal{X}_n$, can be done in time $\text{poly}(\log|\mathcal{X}_n|)$.

Fact 2.3 ([CW81]). For every $t, \ell \in \text{poly}$, there exists an efficient, t -wise independent, function-family ensemble $\{\mathcal{H}_n = \{h: \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}\}\}_{n \in \mathbb{N}}$.

2.3 One-way Functions

Since we deal with “lower bounds”, we focus on *infinitely often*, i.o., one-way functions.¹²

Definition 2.4 (Infinitely often one-way functions (OWF)). A poly-time function $f: \{0, 1\}^* \rightarrow \{0, 1\}^*$ is i.o. one-way if for any PPTM A :

$$\Pr_{\substack{x \leftarrow \{0, 1\}^n \\ y \leftarrow f(x)}} [A(1^n, y) \in f^{-1}(y)] \leq \text{neg}(n) \quad (1)$$

for infinitely many $n \in \mathbb{N}$. The function f is i.o. weakly one-way if for any PPTM A :

$$\Pr_{\substack{x \leftarrow \{0, 1\}^n \\ y \leftarrow f(x)}} [A(1^n, y) \in f^{-1}(y)] \leq 1 - 1/\text{poly}(n) \quad (2)$$

for infinitely many $n \in \mathbb{N}$. The function f is non-uniform i.o. one-way if Equation (1) holds with respect to non-uniform poly-time algorithms (i.e., poly-size circuit family). Non-uniform i.o. weak one-way functions are analogously defined.

We omit 1^n from A ’s input list when clear from the context. When clear from the context, we denote a (length-regular) function $f: \{0, 1\}^* \rightarrow \{0, 1\}^*$ that maps n -bit strings to $p(n)$ -bit strings, by $f: \{0, 1\}^n \rightarrow \{0, 1\}^{p(n)}$.

Definition 2.5 (Strong inversion). A function $f: \{0, 1\}^* \rightarrow \{0, 1\}^*$ is strongly invertible if for every $t \in \text{poly}$ there exists a PPTM A such that:

$$\Pr_{\substack{x \leftarrow \{0, 1\}^n \\ y \leftarrow f(x)}} [A(1^n, y) \in f^{-1}(y)] \geq 1 - 1/t(n).$$

for all $n \in \mathbb{N}$.

Note that if i.o. weak one-way functions do not exist, every poly-time function is strongly invertible.

¹²Since a OWF is also an i.o. OWF, showing limitations on (assumption dependent) black-box reductions from i.o. OWFs to interactive arguments, is a more insightful result.

2.3.1 Weak to Strong One-way Functions

We use the following variant of [Yao82]’s one-way functions amplification, implicit in [Gol01].

Lemma 2.6 (Weak to strong one-way functions). *For every poly-time f_w , $\alpha \in (0, 1)$ and $\varepsilon \geq 1/\text{poly}$, there exist poly-time f and oracle-aided algorithm R such that the following holds: for any algorithm A such that*

$$\Pr_{\substack{x \xleftarrow{R} \{0,1\}^n \\ y \leftarrow f(x)}} [A(1^n, y) \in f^{-1}(y)] > \varepsilon(n)$$

for every $n \in \mathbb{N}$, it holds that

$$\Pr_{\substack{x \xleftarrow{R} \{0,1\}^n \\ y = f_w(x)}} [R^A(1^n, y) \in f_w^{-1}(y)] > \alpha$$

for every $n \in \mathbb{N}$.

Lemma 2.6 holds, under the natural extension, for non-uniform and auxiliary-input one-way functions.

2.4 Interactive Arguments

We use the standard definition of interactive arguments for NP relations.

Definition 2.7 (Interactive argument). *An interactive argument for an NP-relation $\mathcal{R}$ is a pair of PPT algorithms (P, V) such that the following holds:*

Completeness. *For all $(\phi, w) \in \mathcal{R}$: $\Pr[(P(\phi, w), V)(\phi) = 1] = 1$.*¹³

Soundness. *For any PPT $\hat{P}$ and $\phi \notin \mathcal{L}_{\mathcal{R}}$: $\Pr[(\hat{P}, V)(\phi) = 1] \leq \text{neg}(|\phi|)$.*

The prover is ℓ -bit, if for any $(\phi, w) \in \mathcal{R}$, $P(\phi, w)$ sends at most $\ell(|\phi|)$ bits. The interactive argument is laconic, if the prover is $\ell(n) \in O(\log(n))$ -bit. The interactive argument is non uniform, if P and V are non-uniform: receive non-uniform advice z_n for common input of length n .

2.4.1 Zero Knowledge and Witness Hiding

We consider the following strong variants of perfect zero knowledge, witness indistinguishability, and witness hiding interactive arguments. In particular, all definitions below apply to *unbounded* cheating verifiers.¹⁴ Let $\text{View}_{V^*, P(w)}(\phi)$ denote V^* ’s view in a random execution of $(P(w), V^*)(\phi)$.

Definition 2.8 (Perfect zero knowledge). *An interactive argument (P, V) for an NP-relation $\mathcal{R}$ is perfect zero knowledge, if for any probabilistic t -time verifier V^* there exists a probabilistic $\text{poly}(t)$ -time simulator S such that for any $(\phi, w) \in \mathcal{R}$, the distributions $S(\phi)$ and $\text{View}_{V^*, P(w)}(\phi)$ are identical.*

¹³While standard definitions usually allow completeness $1 - \text{neg}(|\phi|)$, requiring perfect completeness only strengthens our results.

¹⁴Since we care about separation results, considering the strongest variants of these definitions is more informative.

Definition 2.9 (Perfect witness indistinguishability). *An interactive argument (P, V) for an NP-relation $\mathcal{R}$ is perfect witness indistinguishable, if for any probabilistic verifier V^* and (ϕ, w_1, w_2) with $(\phi, w_1), (\phi, w_2) \in \mathcal{R}$, the distributions $\text{View}_{V^*, P(w_1)}(\phi)$ and $\text{View}_{V^*, P(w_2)}(\phi)$ are identical.*

Definition 2.10 (Perfect witness hiding). *An interactive argument (P, V) for an NP-relation $\mathcal{R}$ is perfect witness hiding, if for any probabilistic verifier V^* , function h , instance $\phi \in \mathcal{L}(\mathcal{R})$ and distribution W over $\{w : (\phi, w) \in \mathcal{R}\}$, it holds that $\Pr_{w \leftarrow W}[(P(w), V^*)(\phi) = h(w)] \leq 2^{-H_\infty(h(W))}$.*

The proofs of the following facts are immediate.

Fact 2.11. *A perfect zero-knowledge interactive argument for an NP-relation $\mathcal{R}$ is perfect witness hiding for $\mathcal{R}$.*

Fact 2.12. *A perfect witness-hiding interactive argument for an NP-relation $\mathcal{R}$ is perfect witness indistinguishable for $\mathcal{R}$.*

2.5 Games

We capture hardness assumptions using the following notion of games.

Definition 2.13 (Games). *A game G is a pair of a (possibly inefficient) challenger C and a family of sets $\mathcal{I} = \{\mathcal{I}^k = \{i_n^k \in \{0, 1\}^n\}_{n \in \mathbb{N}}\}_{k \in \mathbb{N}}$. Player P wins the game G with probability ε , if for all $k \in \mathbb{N}$, it holds that*

$$\Pr[(P, C)(1^n, i_n^k) = 1] \geq \varepsilon(n)$$

for infinitely many n .

Namely, the player is required to succeed with probability at least ε , for infinitely many n , given any set $\mathcal{I}^k$ in the family. The above definition of games encompasses standard falsifiable assumptions, such as the hardness of inverting one-way functions, but also models more complex assumptions, such as the soundness of an interactive argument, and even assumptions like $\text{NP} \not\subseteq \text{P/poly}$, see below.

2.5.1 Three Games of Interest

We exemplify the above notion of games using three games of interest.

Inverting one-way functions. The following game captures the hardness of inverting a function.

Definition 2.14 (The game G_f^{OWF}). *For a function $f: \{0, 1\}^* \rightarrow \{0, 1\}^*$, the game G_f^{OWF} is the pair $(C, \emptyset)$, where on input 1^n the challenger C samples $x \xleftarrow{R} \{0, 1\}^n$, sends $y \leftarrow f(x)$ to the player, and outputs 1 if the player answers back with $x' \in f^{-1}(y)$.*

It is easy to verify that f is one-way if and only if no PPTM wins the game G_f^{OWF} with non-negligible probability.

Soundness of an interactive argument. The following game captures the soundness of an interactive argument $(\cdot, V)$ with respect to a language $\mathcal{L}$.

Definition 2.15 (The game $G_{V, \mathcal{L}}$). *For a PPTM V and language $\mathcal{L} \subseteq \{0, 1\}^*$, the game $G_{V, \mathcal{L}}$ is the pair $(C, \emptyset)$, where the player first sends $x \in \{0, 1\}^n$, and $C(1^n)$ then run $V(x)$. The challenger C outputs 1 if V accepts the interaction and $x \in \{0, 1\}^n \setminus \mathcal{L}$.*

It is easy to verify that the argument $(\cdot, V)$ is sound for the language $\mathcal{L}$, if and only if no PPTM wins the game $G_{V, \mathcal{L}}$ with non-negligible probability.

Deciding NP. The following game captures the hardness of NP.

Definition 2.16 (The game G^{NP}). *The game G^{NP} is the pair $(C, \mathcal{I})$, where i_n^k is the n -bit binary representation of $\min\{k, 2^n - 1\}$. On input $(1^n, x)$, the player sends $b \in \{0, 1\}$, and the challenger C accepts if and only if $b = 1_{\text{SAT}}(x)$.*

It is easy to verify that the assumption $P \neq \text{NP}$ is equivalent to the assumption that no poly-time player wins the game G^{NP} with probability one. The assumption $\text{NP} \not\subseteq P/\text{poly}$ [resp., $\text{NP} \not\subseteq \text{BPP}$] is equivalent to the assumption that no poly-size [resp., PPT] player wins the game G^{NP} with non-negligible probability.

3 Emulating Assumption-Dependent Reductions to Interactive Arguments

In this section, we define assumption-dependent, fully black-box reductions from one-way functions to interactive arguments and present our main result, showing that the existence of such a reduction implies a reduction from non-uniform one-way functions to the underlying game alone.

In Section 3.1, we formally define assumption-dependent fully black-box reductions from OWFs to succinct interactive arguments. In Section 3.2, we state our main result regarding such reductions. As a warm-up, we prove in Section 3.3 a simplified variant of Theorem 3.2. A full proof of Theorem 3.2 is given in Section 3.4. Finally, in Section 3.5, we explain how to extend our result to reductions from auxiliary-input one-way functions.

3.1 Assumption-Dependent Reductions

In this section, we formally define assumption-dependent, fully black-box reductions from one-way functions to interactive arguments.

Definition 3.1 (Assumption-dependent fully black-box reductions from OWFs to interactive arguments). *An assumption-dependent fully black-box reduction from OWFs to non-uniform, ℓ -bit interactive arguments for an NP-relation $\mathcal{R}$ with respect to a game G , is an oracle-aided, poly-time, non-uniform pair of a function f and a probabilistic algorithm R , such that the following holds for any non-uniform ℓ -bit interactive argument (P, V) for $\mathcal{R}$:*

1. $f^{(P, V)}$ maps n -bit strings to n -bit strings.
2. There exists $p \in \text{poly}$ such that for any $t \in \text{poly}$ and any non-uniform poly-time Inv that inverts $f^{(P, V)}$ with probability $1/t(n)$ for all n , it holds that $R^{(P, V), \text{Inv}}(1^{t(n)}, 1^n)$ wins either the game $G_{V, \mathcal{L}_{\mathcal{R}}}$ or the game G , with probability $1/p(t(n))$.

We make the following remarks:

- Requiring f to be length preserving, which simplifies the notations in our proofs, is not a restriction since an arbitrary one-way function implies a length-preserving one via standard (black-box) techniques.
- The reduction R is black-box in the interactive argument (P, V) , but not in the game G .
- R only has to succeed given access to an inverter, Inv , that inverts f with non-negligible probability for *all* $n \in \mathbb{N}$. Namely, the definition captures reductions from i.o one-way functions.
- R receives $1^{t(n)}$ as an additional input, where $1/t(n)$ is the success probability of Inv . Thus, the definition captures reductions that depend on this parameter.¹⁵
- R has to win one of the games, with probability $1/p(t(n))$. Namely, there should be an *a priori* polynomial relation between the inversion probability of Inv and the winning probability of R . Such a guarantee, however, holds (or can be made to hold) in all reductions we could think of (recall that R gets $1^{t(n)}$ as an additional input).

3.2 Emulating Assumption-Dependent Reductions

In this section, we prove the main result of the paper: an emulation of assumption-dependent, fully black-box reductions from OWFs to interactive arguments.

Theorem 3.2 (Emulating black-box reductions from OWFs to interactive arguments). *Let (f, R) be an assumption-dependent, fully black-box reduction from OWFs to non-laconic,¹⁶ non-uniform, ℓ -bit interactive arguments for an NP-relation $\mathcal{R}$, with respect to a game G . Then there exists a non-uniform, poly-time pair of function g and an oracle-aided randomized algorithm $\hat{R}$, and a constant $c > 0$, such that for any non-uniform poly-time Inv that inverts g with probability $1/\text{poly}(n)$ for all $n \in \mathbb{N}$, it holds that $\hat{R}^{\text{Inv}}$ wins the game G with probability at least c .*

Moreover, the above holds even if the reduction only needs to work when the interactive argument oracle is perfect zero knowledge.

That is, the interactive argument is redundant for the construction of (i.o) one-way functions: if there exists a reduction from one-way functions to *both* interactive arguments and a game G , then there exists a reduction from i.o one-way functions to G alone.

Remark 3.3 (Reductions to $P \neq \text{NP}$). *When applied to (reductions to) the game G^{NP} , see Definition 2.16, Theorem 3.2 yields a reduction from OWFs to the assumption that $\text{NP} \not\subseteq P/\text{poly}$. Indeed, the coins of (and the amplified variant of) algorithm $\hat{R}$ guaranteed by Theorem 3.2, can be fixed so that it wins G^{NP} on all inputs with probability one.*

Remark 3.4 (Reductions to zero-knowledge arguments). *We emphasize that Theorem 3.2 does not address fully black-box reductions from one-way functions to zero-knowledge arguments. In particular, the reduction $\hat{R}$ from Definition 3.1 does not receive oracle access to the zero-knowledge simulator. Rather, the theorem applies to reductions that are only required to work when the argument Π is (perfect) zero knowledge. In particular, this includes reductions that might attempt to exploit the fact that Π is witness indistinguishable or witness hiding.*

¹⁵e.g., the hardcore predicate construction of [GL89].

¹⁶Reductions to laconic interactive arguments are not only uninteresting (laconic arguments have only limited soundness), but they also exist: the reduction breaks the soundness of the argument by sending a uniform string.

Remark 3.5 (Reductions to *non-interactive arguments*). *Theorem 3.2 holds even if we allow the reduction to succeed only when given access to non-interactive arguments. This holds since in the proof, we only use such arguments.*

Remark 3.6 (Infinitely-often inverters). *If we modify Definition 3.1 to work against algorithms that invert the function f (only) infinitely often, then Theorem 3.2 changes accordingly: holds even if we allow the inverter of g to succeed infinitely often.*

3.3 Proof for Length-Preserving Reductions

In this section, we prove Theorem 3.2 for *length-preserving reductions*: reductions such that on input of length n , both f and R query the interactive argument algorithms on statements of length n , and R queries the inverter of f on inputs of length n . Let (f, R) be a length-preserving, assumption-dependent, fully black-box reduction from OWFs to non-uniform ℓ -bit interactive arguments for the NP-relation $\mathcal{R}$, with respect to a game G , where $\ell(n) \in \omega(\log n)$ (recall the reduction is to non-laconic arguments). In the following, we fix the first input of R , $1^{t(n)}$, to 1^8 (i.e., R fixed to work against inverters with success probability at least $1/8$), and omit it from the notation. Throughout, we assume that f only makes well-formed, valid prover-queries, i.e., $(\phi, w) \in \mathcal{R}$ (by design, our prover returns $\perp$ on invalid queries). Let $p_f(n) \in \text{poly}(n)$ be an upper bound on the total running time of f and R , on inputs of length n .

3.3.1 The Interactive Argument

We start by defining a non-interactive argument Π to which the reduction will get oracle access. Towards this goal, we consider the following non-uniform, non-interactive argument, defined using an efficient $p_f(n)$ -wise independent, function-family ensemble $\mathcal{H} = \{\mathcal{H}_n = \{h: \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}\}\}_{n \in \mathbb{N}}$.

Algorithm 3.7. [Hash-based argument $\hat{\Pi} = (\hat{P}, \hat{V})$]

Parameter: $h \in \mathcal{H}_n$.

$\hat{P}_h(\phi, w)$: If $(\phi, w) \in \mathcal{R}$, return $h(\phi)$; otherwise return $\perp$.

$\hat{V}_h(\phi, \pi)$: If $\pi = h(\phi)$, return 1; otherwise return 0.

That is, the prover simply applies the hash on the statement (if an appropriate witness is provided), and the verifier checks if the proof matches the hash of the statement. Algorithm 3.7 is not the argument on which we invoke R , but will be used as an intermediate tool for characterizing the actual argument. In particular, we use the following “query function” qry that outputs the prover queries made by $f^{\hat{\Pi}}$.

Function 3.8. [The query function qry] On input $(h, x, i) \in \mathcal{H}_n \times \{0, 1\}^n \times [p_f(n)]$, the function qry executes $f^{\hat{\Pi}_h}(x)$ up to the i^{th} prover-query (ϕ_i, w_i) , and outputs ϕ_i . If $f^{\hat{\Pi}_h}(x)$ does not make i prover queries, it outputs $\perp$.

A key observation is that without loss of generality $\mathbf{qry}$ (defined for any value of n) is strongly invertible. Otherwise, there exists a black-box reduction from the inversion of $\mathbf{qry}$ to $\mathbf{G}$, and we are done.¹⁷ Consequently, there exists a poly-time, non-uniform algorithm $\mathbf{stInv}$ that inverts $\mathbf{qry}$ with probability $1 - \varepsilon(n)$ (over the uniform input distribution), for $\varepsilon(n) := 1/4p_f(n)$. We use $\mathbf{stInv}$ to construct the non-interactive argument $\Pi = (\mathbf{P}, \mathbf{V})$. The non-uniform prover and verifier are parameterized by two hash functions: $h^{\mathbf{C}}$, for “easy” instances, where the verifier checks for a witness, and $h^{\mathbf{T}}$, for “hard” instances, where the verifier must trust the proof.

Algorithm 3.9. [Prover $\mathbf{P}_{h^{\mathbf{C}}, h^{\mathbf{T}}}$]

Parameters: $h^{\mathbf{C}}, h^{\mathbf{T}} \in \mathcal{H}_n$.

Input: $(\phi, w) \in \{0, 1\}^n \times \{0, 1\}^*$

Operation:

1. If $(\phi, w) \notin \mathcal{R}$, return $\perp$.
2. Let $z \leftarrow \mathbf{stInv}(\phi)$.
3. Return $h^{\mathbf{C}}(\phi)$ if $\mathbf{qry}(z) = \phi$, and $h^{\mathbf{T}}(\phi)$ otherwise.

That is, the prover applies $h^{\mathbf{C}}$ on the statement if $\mathbf{stInv}$ succeeds in inverting it, and applies $h^{\mathbf{T}}$ otherwise.

Algorithm 3.10. [Verifier $\mathbf{V}_{h^{\mathbf{C}}, h^{\mathbf{T}}}$]

Parameters: $h^{\mathbf{C}}, h^{\mathbf{T}} \in \mathcal{H}_n$.

Input: $(\phi, \pi) \in \{0, 1\}^n \times \{0, 1\}^*$

Operation:

1. If $\pi = h^{\mathbf{T}}(\phi)$: return 1.
2. If $\pi = h^{\mathbf{C}}(\phi)$:
 - (a) Let $z \leftarrow \mathbf{stInv}(\phi)$.
 - (b) If $\mathbf{qry}(z) = \phi$, return 1.
3. Return 0.

That is, if the proof π matches $h^{\mathbf{T}}(\phi)$, then the verifier accepts. If the proof matches $h^{\mathbf{C}}(\phi)$, the verifier runs $\mathbf{stInv}$, and if succeeds, which implies that $\phi \in \mathcal{L}_{\mathcal{R}}$, it accepts. For the non-uniform, non-interactive argument that is defined over *all* input lengths, we take $\Pi = \{(\mathbf{P}_{h_n^{\mathbf{C}}, h_n^{\mathbf{T}}}, \mathbf{V}_{h_n^{\mathbf{C}}, h_n^{\mathbf{T}}})\}_{(h_n^{\mathbf{C}}, h_n^{\mathbf{T}}) \in \mathcal{H}_n}$ (the choice of hash functions will be discussed later). For most parts, however, it will be sufficient to consider a single input length. Since $\mathbf{stInv}$ is deterministic, its executions by the prover and

¹⁷If a function $\mathbf{qry}$ is not strongly invertible, by Lemma 2.6 it can be amplified to a one-way function g from which a security reduction can be defined to *any* game $\mathbf{G}$; any efficient inverter of g breaks the hardness of $\mathbf{G}$ (since such inverter does not exist).

the verifier are identical. Hence, Π has *perfect* completeness. Also note that Π is a perfect zero-knowledge argument: on input ϕ , the simulator acts according to $P(\phi, \cdot)$ without the first step (i.e., not testing for $(\phi, w) \in \mathcal{R}$).

Towards presenting the (candidate one-way) function g and the reduction $\widehat{R}$, we prove that the executions of $f^{\widehat{\Pi}}$ and f^{Π} are close.

Claim 3.11. $\Pr_{(h^C, h^T, x) \xleftarrow{R} \mathcal{H}_n^2 \times \{0,1\}^n} [f^{\widehat{\Pi}_{h^C}}(x) \neq f^{\Pi_{h^C, h^T}}(x)] \leq 1/2$ for all n with $\ell(n) \geq 2 + \log 1/\varepsilon(n)$.

Proof. We prove the stronger statement that the *execution traces*—the input x and the inputs/outputs of the oracle calls—of $f^{\Pi_{h^C, h^T}}(x)$ and $f^{\widehat{\Pi}_{h^C}}(x)$ are $1/2$ close, for uniform (x, h^C, h^T) . The proof is by coupling the two distributions.

Algorithm 3.12. [Coupling f^{Π} and $f^{\widehat{\Pi}}$]

1. Sample $(h^C, h^T) \xleftarrow{R} \mathcal{H}_n^2$ and $x \xleftarrow{R} \{0,1\}^n$.
2. Let $\text{trace}^{\widehat{\Pi}}$ and trace^{Π} denote the traces of the executions of $f^{\widehat{\Pi}_{h^C}}(x)$ and $f^{\Pi_{h^C, h^T}}(x)$, respectively.
3. Output $(\text{trace}^{\widehat{\Pi}}, \text{trace}^{\Pi})$.

We prove the claim by showing that

$$\Pr[\text{trace}^{\widehat{\Pi}} \neq \text{trace}^{\Pi}] \leq 1/2 \quad (3)$$

Let E_i be the event that $\text{trace}^{\widehat{\Pi}}$ and trace^{Π} are identical up to the i^{th} query. We show that $\Pr[\neg E_i \mid E_{i-1}] \leq 2\varepsilon(n)$, for every $i \in [p_f(n)]$, and since E_0 trivially holds, the proof of Equation (3) follows by a union bound. Assuming E_{i-1} holds, the i^{th} query generated by f is the *same* in both executions. We bound the probability that the oracle responses on it differ (i.e., that $\neg E_i$ occurs), while separately handling prover and verifier queries. Since the reduction is length preserving, all queries are of the form $(\phi, \cdot) \in \{0,1\}^n \times \{0,1\}^*$.

Prover query. Let (ϕ, w) be a query to the prover. By our assumption on f , we have $(\phi, w) \in \mathcal{R}$.

By definition, $\widehat{P}_{h^C}$ answers $h^C(\phi)$. The response of P_{h^C, h^T} depends on the output of $\text{stInv}(\phi)$.

Since ϕ is distributed as a uniform query of $f^{\widehat{\Pi}}$ conditioned on E_{i-1} , which by the induction assumption occurs with probability at least $1/2$, it holds that stInv outputs a preimage z with probability at least $1 - 2\varepsilon(n)$ (over the choice of h^C, x). If this happens, then P also answers $h^C(\phi)$.

Verifier query. Let (ϕ, π) be a query to the verifier. Assume π was generated by a previous prover query. Since E_{i-1} holds, we have $\pi = h^C(\phi)$. Thus, stInv succeeds in inverting ϕ , and both V_{h^C, h^T} and $\widehat{V}_{h^C}$ accept.

Assume π was not generated by a previous prover query. Since $\mathcal{H}$ is a p_f -wise independent family, $\widehat{V}$ accepts with probability $2^{-\ell(n)}$, and V accepts with probability $\in [2^{-\ell(n)}, 2 \cdot 2^{-\ell(n)}]$. Hence, they answer the same value with probability at least $1 - 2^{1-\ell(n)}$, which is at least $1 - 2\varepsilon(n)$ for the considered n .

Combining the two cases, we deduce that $\Pr[\neg E_i \mid E_{i-1}] \leq 2\varepsilon(n)$, concluding the proof. $\square$

3.3.2 The Function g

The function g , whose security is reduced to the game G , is defined using the function $g_w: \mathcal{H}_n \times \{0, 1\}^n \rightarrow \mathcal{H}_n \times \{0, 1\}^n$, defined by:

$$g_w(h, x) = (h, f^{\hat{\Pi}_h}(x)). \quad (4)$$

Let g and R_{amp} be the (efficient) non-uniform function and reduction guaranteed by Lemma 2.6, such that for any efficient non-uniform A that inverts g with non-negligible probability, it holds that R_{amp}^A inverts g_w with probability at least $3/4$.

3.3.3 The Reduction $\hat{R}$

The black-box reduction from the one-wayness of g to the hardness of G is defined as follows.

Algorithm 3.13. $[\hat{A}_h]$

Oracle: A .

Parameter: $h \in \mathcal{H}_n$.

Input: $y \in \{0, 1\}^n$.

Operation: Output $R_{\text{amp}}^A(h, y)_2$, i.e., the second output of R_{amp}^A .

Algorithm 3.14. [Reduction $\hat{R}$ from g to G]

Oracle: A .

Input: 1^n .

Operation:

1. Sample $(h^C, h^T) \xleftarrow{R} \mathcal{H}_n^2$.
2. Act in G according to $R^{\hat{A}_{h^C}, \Pi_{h^C, h^T}}(1^n)$.

We prove that $\hat{R}$ wins G with constant probability. Let p be the polynomial from Definition 3.1 with respect to R .

Claim 3.15. *For any non-uniform poly-time A that inverts g with probability $1/\text{poly}$ for all $n \in \mathbb{N}$, it holds that $\hat{R}^A$ wins G with probability $1/9p(8)$.*

Proof. Let A be a non-uniform poly-time algorithm that inverts g with probability $1/\text{poly}$ for all $n \in \mathbb{N}$. By Lemma 2.6, R_{amp}^A inverts g_w with probability at least $3/4$. Let $(g_w)_y(h, x) := g_w(h, x)_2$, i.e., it outputs the second output of g_w . By the definition of $\hat{A}$,

$$\Pr_{(h^C, x) \xleftarrow{R} \mathcal{H}_n \times \{0, 1\}^n; y \leftarrow (g_w)_y(h^C, x)} \left[(g_w)_y(h^C, \hat{A}_{h^C}(y)) = y \right] \geq 3/4.$$

Hence, by Claim 3.11

$$\Pr_{(h^C, h^T, x) \leftarrow \mathcal{H}_n^2 \times \{0,1\}^n; y \leftarrow f^{\Pi_{h^C, h^T}}(x)} \left[f^{\Pi_{h^C, h^T}}(\hat{A}_{h^C}(y)) = y \right] \geq 1/4 \quad (5)$$

for all n with $\ell(n) \geq 2 + \log 1/\varepsilon(n)$. Since we can modify $\hat{A}$ to invert g_w with probability one on all small n , in the following we assume without loss of generality that Equation (5) holds for *all* $n \in \mathbb{N}$. Applying a Markov argument to Equation (5), we get that

$$\Pr_{d_n \leftarrow (h^C, h^T) \leftarrow \mathcal{H}_n^2} \left[\underbrace{\Pr_{x \leftarrow \{0,1\}^n; y \leftarrow f^{\Pi_{d_n}}(x)} \left[f^{\Pi_{d_n}}(\hat{A}_{h^C}(y)) = y \right] \geq 1/8}_{\delta(d_n)} \right] \geq 1/8 \quad (6)$$

Let $(C_G, \mathcal{I})$ be the challenger and the family of sets of the game G , respectively. For $n \in \mathbb{N}$ and $(h^C, h^T) \in \mathcal{H}_n^2$, let C_{h^C, h^T} be the (possibly inefficient) algorithm that on common input 1^n , acts according to V_{h^C, h^T} and outputs 1 if V accepts a statement $x \in \{0,1\}^n \setminus \mathcal{L}_{\mathcal{R}}$ (i.e., the challenger of the game $G_{V, \mathcal{L}_{\mathcal{R}}}$ when defined only for input length n). We claim that for all $k \in \mathbb{N}$ and infinitely many $n \in \mathbb{N}$:

$$\Pr_{d_n \leftarrow (h^C, h^T) \leftarrow \mathcal{H}_n^2} \left[\underbrace{\Pr \left[(R^{\hat{A}_{h^C, \Pi_{d_n}}, C_{d_n}})(1^n) = 1 \vee (R^{\hat{A}_{h^C, \Pi_{d_n}}, C_G})(1^n, i_n^k) = 1 \right] \geq 1/p(8)}_{\rho(d_n)} \right] \geq 1/8 \quad (7)$$

where the inner probability is over the randomness of R . Indeed, assume Equation (7) does not hold. Hence, by Equation (6),

$$\Pr_{d_n} [\delta(d_n) \wedge \neg \rho(d_n)] > 0$$

for all but finitely many n . Hence, there exists $\{d_n \leftarrow (h_n^C, h_n^T) \in \mathcal{H}_n^2\}_{n \in \mathbb{N}}$ such that $\delta(d_n) = 1$ and $\rho(d_n) = 0$ for all but finitely many n . It follows that there exist a non-uniform interactive argument Π that gets $\{d_n\}_{n \in \mathbb{N}}$ as a non-uniform advice, and a non-uniform algorithm $\hat{A}$ that gets $\{h_n^C\}_{n \in \mathbb{N}}$ as a non-uniform advice, such that the following holds:

- $\hat{A}$ inverts f^{Π} with probability at least $1/8$, for all n .
- $R^{\hat{A}, \Pi}$ does *not* win $G_{V, \mathcal{L}_{\mathcal{R}}}$ nor G , with probability $1/p(8)$.

This, however, contradicts the definition of R , and we conclude that Equation (7) holds.

We next claim that for any large enough $n \in \mathbb{N}$, and any $h^C \in \mathcal{H}_n$, it holds that

$$\Pr_{h^T \leftarrow \mathcal{H}_n; d_n \leftarrow (h^C, h^T)} \left[(R^{\hat{A}_{h^C, \Pi_{d_n}}, C_{d_n}})(1^n) = 1 \right] \leq \text{neg}(n) \quad (8)$$

Indeed, since A was fixed before h^T was sampled, and since g is independent of h^T , the calls to $\hat{A}_{h^C}$ reveal *no information* regarding h^T . Since in addition R makes fewer than $p_f(n)$ calls to Π_{h^C, h^T} , the probability it generates an accepting proof of the form $h^T(\phi)$ for a statement ϕ of length n that was not returned by P_{h^C, h^T} is negligible, i.e., $2^{-\ell(n)}$. We also note that, by design, V_{h^C, h^T}

does not accept instances $x \notin \mathcal{L}_{\mathcal{R}}$ with a proof of the form $h^{\mathcal{C}}(x)$ (unless $h^{\mathcal{C}}(x) = h^{\mathcal{T}}(x)$, which occurs with negligible probability).

Combining Equations (7) and (8), we conclude that for all $k \in \mathbb{N}$ and infinitely many $n \in \mathbb{N}$:

$$\Pr_{d_n \leftarrow (h^{\mathcal{C}}, h^{\mathcal{T}}) \leftarrow \mathcal{H}_n^2} \left[(\mathbf{R}^{\hat{A}_{h^{\mathcal{C}}, \Pi_{d_n}}}, \mathbf{C}_{\mathbf{G}})(1^n, i_n^k) = 1 \right] \geq 1/(8p(8)) - \text{neg}(n).$$

Namely, $\hat{R}^A$ wins $\mathbf{G}$ with probability (at least) $1/(9p(8))$. $\square$

3.3.4 Putting it Together

Proving the length-preserving variant of Theorem 3.2. The proof for length-preserving reductions follows directly from Claim 3.15. $\square$

3.4 Proof for Arbitrary Reductions

In this section, we prove Theorem 3.2 for arbitrary reductions. Let $(f, \mathbf{R})$ be an assumption-dependent, fully black-box reduction from OWFs to ℓ -bit non-uniform interactive arguments for the NP-relation $\mathcal{R}$, with respect to a game $\mathbf{G}$, where $\ell(n) \in \omega(\log n)$. As in the length-preserving case, we fix the first input of $\mathbf{R}$, $1^{t(n)}$, to 1^8 , and omit it from the notation, and assume that f only makes well-formed, valid prover-queries, i.e., $(\phi, w) \in \mathcal{R}$. In addition, we assume weak i.o non-uniform one-way functions do not exist.¹⁸ Let $p_f(n) \in \text{poly}$ bound the running time of f and $\mathbf{R}$, on inputs of length n . We assume without loss of generality that $p_f(n)$ is monotonically increasing, and thus the function $p_f^{-1}(n)$ is well defined.

3.4.1 The Interactive Argument

Let $\mathcal{H} = \{\mathcal{H}_n = \{h: \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}\}\}_{n \in \mathbb{N}}$ be an efficient $p_f(p_I(n))$ -wise independent, function-family ensemble, where $p_I \in \text{poly}$ will be determined by the analysis. The non-interactive argument $\Pi = (\mathbf{P}, \mathbf{V})$, to which f and $\mathbf{R}$ will get oracle access, is defined next, using the function `qry` and algorithm `stInv` to be defined afterwards.

¹⁸In the length-preserving proof, we assumed that the query function `qry`, defined by the reduction, is not one-way. The very same argument, however, applies for *any* efficient function: if a one-way function g exists, then the trivial black-box reduction that does nothing is a black-box reduction from g to $\mathbf{G}$.

Algorithm 3.16. [Prover P_{h^C, h^T}]Parameters: $h^C, h^T \in \mathcal{H}_n$.Input: $(\phi, w) \in \{0, 1\}^n \times \{0, 1\}^*$.

Operation:

1. If $(\phi, w) \notin \mathcal{R}$, return $\perp$.
2. For $\lambda = \lceil p_f^{-1}(n) \rceil$ to $p_f(n)$:
 - (a) Let $z \leftarrow \text{stInv}(1^\lambda, \phi)$.
 - (b) If $\text{qry}(z) = \phi$, return $h^C(\phi)$.
3. Return $h^T(\phi)$.

Algorithm 3.17. [Verifier V_{h^C, h^T}]Parameters: $h^C, h^T \in \mathcal{H}_n$.Input: $(\phi, \pi) \in \{0, 1\}^n \times \{0, 1\}^*$

Operation:

1. If $\pi = h^T(\phi)$, return 1.
2. If $\pi = h^C(\phi)$, for $\lambda = \lceil p_f^{-1}(n) \rceil$ to $p_f(n)$:
 - (a) Let $z \leftarrow \text{stInv}(1^\lambda, \phi)$.
 - (b) If $\text{qry}(z) = \phi$, return 1.
3. Return 0.

That is, the argument is similar to that of the length-preserving proof, but uses a more complicated query function qry , defined next, and a strong inverter stInv of qry . In addition, when it calls stInv , it does so on *different* input lengths. This modification comes to handle a function f that calls Π on instances of length different than its input length. Recall that in the length-preserving proof we effectively prevented f from accessing the hash function h^T . Here, the above protocol only enables it to access h^T on short input, which suffices for retaining the soundness of the interactive argument against R . For the non-uniform, non-interactive argument that is defined over *all* input lengths, we take $\Pi = \{(P_{h_n^C, h_n^T}, V_{h_n^C, h_n^T})\}_{(h_n^C, h_n^T) \in \mathcal{H}_n}$ (the choice of hash functions will be discussed later). As in the length-preserving proof, it is easy to see that Π has perfect completeness, and is perfect zero knowledge.

The query function qry . To define the query function qry , we use the following notation: for $\mathbf{h} = (h_1, h_2, \dots, h_n) \in \mathbf{X}_{j=1}^n \mathcal{H}_j = \mathcal{H}_1 \times \dots \times \mathcal{H}_n$, $k \in [n]$ and $x \in \{0, 1\}^k$, let $\mathbf{h}(x) := (\mathbf{h})_{|x|}(x)$.

Function 3.18. [Query function **qry**]

Input: $(i, x, \mathbf{h}^C, \mathbf{h}^\top) \in [p_f(n)] \times \{0, 1\}^n \times \prod_{j=1}^{p_f(n)} \mathcal{H}_j \times \prod_{j=1}^{\lceil p_I^{-1}(n) \rceil - 1} \mathcal{H}_j$.

Operation:

1. Emulate $f(x)$ while answering its oracle queries as follows:

Prover query (ϕ, w) :

- (a) If $(\phi, w) \notin \mathcal{R}$, return $\perp$.
- (b) If $|\phi| \geq p_I^{-1}(n)$, return $\mathbf{h}^C(\phi)$. Else, return $P_{(\mathbf{h}^C)_{|\phi|}, (\mathbf{h}^\top)_{|\phi|}}(\phi, w)$.

Verifier query (ϕ, π) : If $|\phi| \geq p_I^{-1}(n)$, return 1 if and only if $\pi = \mathbf{h}^C(\phi)$. Else, return $V_{(\mathbf{h}^C)_{|\phi|}, (\mathbf{h}^\top)_{|\phi|}}(\phi, \pi)$.

2. If the execution reaches the i^{th} prover query (ϕ_i, w_i) , return ϕ_i . Else, return $\perp$.

That is, on “not too short statements”, the query function only uses the hash function $\mathbf{h}^C$ (i.e., as in the length-preserving proof). On short statements, it acts according to Π .

The strong inverter stInv and the polynomial p_I . We define the inversion algorithm stInv , used above, and argue about the efficiency of the query function **qry** (and thus of Π). We also address here the polynomial p_I , used above to parameterize the construction.

Defining the efficient strong inverter stInv of the query function **qry** should be done with care since **qry** calls stInv (on shorter inputs). Yet, below we prove that such a strong inverter exists.

Let $t(n) := p_f(n) + n + \sum_{j=1}^{p_f(n)} |h_j| + \sum_{j=1}^{\gamma(n)} |h_j|$, where for all j , $h_j \in \mathcal{H}_j$ and $\gamma(n) := \lceil p_I^{-1}(n) \rceil - 1$. Let $\mathbb{N}_q := \{t(n) : n \in \mathbb{N}\}$, i.e., all input lengths of **qry**. Let $\varepsilon(m) := 1/p_f^3(m)$.

Claim 3.19 (Existence of a strong inverter for **qry**). *There exists $p_I \in \text{poly}$ and a poly-time, non-uniform algorithm stInv such that*

$$\Pr_{z \leftarrow \{0,1\}^m} [\text{stInv}(\text{qry}(z)) \in \text{qry}^{-1}(\text{qry}(z))] \geq 1 - \varepsilon(m)$$

for every $m \in \mathbb{N}_q$.

Claim 3.19 immediately yields that **qry** is also poly-time non-uniform computable. To prove Claim 3.19, we use the following lemma, proved in Section 3.4.7, which gives good control on the inversion time of an efficient function.

Lemma 3.20 (Universal inversion time for non-uniform functions). *Assume that i.o non-uniform one-way functions do not exist. Then for every $p_E \in \text{poly}$ and $\varepsilon(n) \geq 1/\text{poly}(n)$, there exists $p_I \in \text{poly}$ such that the following holds: for every circuit f_n over $\{0,1\}^n$ of size $p_E(n)$, there exists a $p_I(n)$ -size circuit C_n such that $\Pr_{x \leftarrow \{0,1\}^n} [C_n(f_n(x)) \in f_n^{-1}(f_n(x))] \geq 1 - \varepsilon(n)$.*

Namely, the size of a strong inverter of a $p_E(n)$ -size f , is bounded by a (universal) polynomial in $p_E(n)$.

Proof of Claim 3.19. Let $p_I \in \text{poly}$ be the (smallest) polynomial guaranteed by Lemma 3.20 with respect to $p_E(m) := p_f^7(m)$ and $\varepsilon(m)$. We define stInv as a (poly-size) circuit family $\{\text{stInv}_j\}_{j \in \mathbb{N}}$.¹⁹ Conveniently, we also view the query function qry as a circuit family $\{\text{qry}_m\}_{m \in \mathbb{N}_q}$.

We prove by induction on $m \in \mathbb{N}_q$, that $|\text{qry}_m| \leq p_E(m)$, while defining the strong inverter stInv along the way. We assume that qry_m makes $p_f(t^{-1}(m))$ oracle queries, an over-counting that allows us to ignore the computation cost of the parts of qry_m that are not oracle queries. The base case is clear. Proving for $m \in \mathbb{N}_q$, the induction hypothesis yields that $|\text{qry}_{m'}| \leq p_E(m')$ for all $m' \in \mathbb{N}_q$ with $m' < m$. For any such m' , let $C_{m'}$ be the $p_I(m')$ -size circuit guaranteed by Lemma 3.20, that inverts $\text{qry}_{m'}$ on inputs of length m' . For $j \in \mathbb{N}$ with $t(j) < m$, let $\text{stInv}_j(\phi) := C_{t(j)}(\phi)$. We prove that

$$|\text{qry}_m| \leq p_E(m) \quad (9)$$

To see the above, consider an execution of the algorithm qry on an input of length $m = t(n)$. The execution emulates $f(x)$ for $x \in \{0, 1\}^n$, up to the i^{th} oracle query, for $i \in [p_f(n)]$. In the following we bound the running time of each query $(\phi, \cdot)$. Let $k \leftarrow |\phi|$.

Case $k \geq p_I^{-1}(n)$. We assume without loss of generality that the polynomial that bounds the evaluation time of the hash functions $\mathcal{H}$ (see Section 2.2) is bounded by $p_f^2(p_I(n))$ (If this is not the case, we would consider larger $p_f \in \text{poly}$). Since qry computes $\mathbf{h}^C(\phi)$ in this case, and since $k \leq p_f(n)$, the running time of this query is bounded by $p_f^2(p_I(p_f(n)))$.

Finally, since $t(n) > p_I(p_f(n))$ (recall that $t(n)$ also accounts for the description of $h \in \mathcal{H}_{p_f(n)}$), we conclude that $p_f^2(p_I(p_f(n))) \leq p_f^2(t(n)) = p_f^2(m)$.

Case $k < p_I^{-1}(n)$. On such queries, qry invokes Π which in turn tries to invert ϕ using $\text{stInv}(1^\lambda, \phi)$, for every $\lambda \in [[p_f^{-1}(k)], p_f(k)]$. Since $m = t(n)$ and assuming without loss of generality that $p_I > p_f$, we deduce that for $k < p_I^{-1}(n)$ it holds that $t(p_f(k)) < m$. Thus, we can apply the induction hypothesis to deduce that qry can be computed in time $p_E(t(p_f(k)))$. Hence, by Lemma 3.20, the computation size (and running time) of each inversion attempt is bounded by $p_I(t(p_f(k))) \leq t(p_f(n)) \leq p_f(m)$.²⁰ Finally, since Π makes at most $p_f(k)$ inversion attempts, the total emulation time is bounded by $p_f(k) \cdot p_f(m) \leq p_f(m)^2 \leq p_f^2(m)$.²¹

Since qry makes at most $p_f(n)$ queries, its total running time is bounded by $p_f^3(m)$. Since we can assume without loss of generality that the size of the non-uniform advice of f on input length n (and thus of qry on input length m) is bounded by $p_f(n)$, and that the description of the algorithm f is included in its non-uniform advice, we conclude that qry_m can be computed by a circuit of size $(p_f^3(m) + p_f(n))^2 < p_E(m)$.

Since the above also defines the circuit family $\{\text{stInv}_j\}_{j \in \mathbb{N}}$ of size $p_I(t(j)) \in \text{poly}(j)$, and proves that it inverts qry with probability $1 - \varepsilon(m)$ for all $m \in \mathbb{N}_q$, this concludes the proof. $\square$

¹⁹Clearly, for such a family there exists a poly-time, non-uniform algorithm that computes it, as required by the claim.

²⁰While polynomial composition is not commutative for arbitrary polynomials, it does hold in our case since without loss of generality the leading coefficients of t and p_f are one.

²¹This holds since without loss of generality p_f is of degree at least two.

3.4.2 The Query Function of f^Π

A key observation is that the above “query function” $\mathbf{qry}$ emulates well the (real) query function induced by f^Π —the queries f^Π makes to the oracles in a random execution. (We later use this fact to deduce that an inverter for $\mathbf{qry}$ is also an inverter for the query function induced by f^Π .)

As in the length-preserving proof, we consider the function f_{trace}^Π that on input $x \in \{0,1\}^*$, outputs x and all oracle queries/answers made by $f^\Pi(x)$. We compare it to the function $\mathbf{qry}_{\text{trace}}$ that on input $(x, \mathbf{h}^C, \mathbf{h}^\top)$, outputs x and the oracle queries/answers made by the emulated execution of f in the execution of $\mathbf{qry}(p_f(|x|), x, \mathbf{h}^C, \mathbf{h}^\top)$. Specifically, we show that the output distributions of these transcript functions are statistically close.

Claim 3.21 (Similarity between f_{trace}^Π and $\mathbf{qry}_{\text{trace}}$). *For every $n \in \mathbb{N}$:*

$\text{SD}(f_{\text{trace}}^{\Pi_{H^C, H^\top}}(X), \mathbf{qry}_{\text{trace}}(X, H^C, \widetilde{H^\top})) \leq p_f(n) \cdot 2^{-\ell(p_I^{-1}(n))}$, where $X \leftarrow U_n$, $(H^C, H^\top) \xleftarrow{R} (\bigtimes_{j=1}^{p_f(n)} \mathcal{H}_j)^2$ and $\widetilde{H^\top} \leftarrow (H^\top)_{\leq [p_I^{-1}(n)]-1}$.

Proof. We prove the claim by presenting a random process, parameterized by n , whose output distribution is identical to that of $\mathbf{qry}_{\text{trace}}$, and statistically close to that of f_{trace}^Π .

Algorithm 3.22.

1. Sample $(x, \mathbf{h}^C, \mathbf{h}^\top) \xleftarrow{R} \{0,1\}^n \times (\bigtimes_{j=1}^{[p_I^{-1}(n)]-1} \mathcal{H}_j)^2$.
2. Sample $a_1, \dots, a_{p_f(n)} \xleftarrow{R} \{0,1\}^{\ell(p_f(n))}$.
3. Execute $f(x)$ while answering its i^{th} oracle query (ϕ, z) as follows:
 - Short queries** $|\phi| \leq [p_I^{-1}(n)] - 1$: Return according to $\Pi_{(\mathbf{h}^C)_{|\phi|}, (\mathbf{h}^\top)_{|\phi|}}$.
 - Long queries** $|\phi| > [p_I^{-1}(n)] - 1$:
 - Prover query:** Return $(a_i)_{i \leq \ell(|\phi|)}$.
 - Verifier query:** If z was an answer to a previous prover query $(\phi, \cdot)$, accept. Otherwise, accept with probability $2^{-\ell(|\phi|)}$.
4. Output the transcript of the execution.

The only difference between $\mathbf{qry}_{\text{trace}}$ and f_{trace}^Π is the way long queries are answered; in $\mathbf{qry}_{\text{trace}}$, it is (always) according to $\mathbf{h}^C$. In f_{trace}^Π , it is according to $\mathbf{h}^C$ if the inversion succeeds, and according to $\mathbf{h}^\top$ otherwise. We separately analyze prover and verifier queries.

Prover queries. Since the number of queries made by f (and thus by $\mathbf{qry}$) is smaller than the many-wise independence of the hash functions used for such long queries ($|\phi| > p_I^{-1}(n)$, hence $p_f(p_I(|\phi|)) > p_f(n)$), in both cases the hash functions’ answers are uniform values, which is exactly what the prover replies in the above random process.

Verifier queries. In both cases the verifier accepts if the proof π is consistent with the value that is generated by the prover. In the case where the verifier query was not generated by a prior prover query, in $\mathbf{qry}_{\text{trace}}$ the verifier accepts only if the proof matches $\mathbf{h}^C$, i.e., with probability $2^{-\ell(|\phi|)}$, as in the random process. In f_{trace}^Π , the verifier accepts if the proof matches $\mathbf{h}^\top$, and

might also accept if the proof matches $\mathbf{h}^C$. The probability that the proof matches at least one of the hashes is at most $2 \cdot 2^{-\ell(|\phi|)}$. Since the number of queries is bounded by $p_f(n)$ and the length of the statement in such verifier queries is at least $p_I^{-1}(n)$, by a union bound, the statistical difference between the distributions is at most $p_f(n) \cdot 2^{-\ell(p_I^{-1}(n))}$. $\square$

3.4.3 The Punctured Argument

Towards defining the function g , whose one-wayness can be reduced to the game G , we define the “punctured” variant Π^{punct} of Π such that: (1) Inverting a function with oracle access to Π^{punct} is not useful for breaking the security of Π . (2) The execution of $f^{\Pi^{\text{punct}}}$ is close to that of f^Π .²² The above is achieved by removing the access to the hash function $h^\top$ on the relevant input lengths (i.e., we “punctured” Π). Let $\text{tr}(n) := \lceil p_f^{-1}(n) \rceil - 1$.

Algorithm 3.23. [Prover $\mathsf{P}_{\mathbf{h}^C, \mathbf{h}^\top}^{\text{punct}}$]

Parameters: $\mathbf{h}^C \in \mathsf{X}_{j=1}^{p_f(n)} \mathcal{H}_j$, $\mathbf{h}^\top \in \mathsf{X}_{j=1}^{\text{tr}(n)} \mathcal{H}_j$.

Input: $(\phi, w) \in (\{0, 1\}^*)^2$.

Operation:

1. If $(\phi, w) \notin \mathcal{R}$ or $|\phi| > p_f(n)$, return $\perp$.
2. If $|\phi| > \text{tr}(n)$, return $\mathbf{h}^C(\phi)$.
3. Return $\mathsf{P}_{(\mathbf{h}^C)_{|\phi|}, (\mathbf{h}^\top)_{|\phi|}}(\phi, w)$.

Algorithm 3.24. [Verifier $\mathsf{V}_{\mathbf{h}^C, \mathbf{h}^\top}^{\text{punct}}$]

Parameters: $\mathbf{h}^C \in \mathsf{X}_{j=1}^{p_f(n)} \mathcal{H}_j$, $\mathbf{h}^\top \in \mathsf{X}_{j=1}^{\text{tr}(n)} \mathcal{H}_j$.

Input: $(\phi, \pi) \in (\{0, 1\}^*)^2$.

Operation:

1. If $|\phi| > p_f(n)$, return 0.
2. If $|\phi| > \text{tr}(n)$: return 1 if $\pi = \mathbf{h}^C(\phi)$ and 0 otherwise.
3. Return $\mathsf{V}_{(\mathbf{h}^C)_{|\phi|}, (\mathbf{h}^\top)_{|\phi|}}(\phi, \pi)$.

That is, Π^{punct} operates similarly to Π , but the parties use *only* $\mathbf{h}^C$ for long statements. We prove that the executions of f^Π and $f^{\Pi^{\text{punct}}}$ are similar (analogous to Claim 3.11 in the length-preserving proof).

For $\mathbf{h}^C, \mathbf{h}^\top \in (\mathsf{X}_{j=1}^n \mathcal{H}_j)^2$, let $\Pi_{\mathbf{h}^C, \mathbf{h}^\top}$ be the argument that on input $(\phi, \cdot) \in \{0, 1\}^k \times \{0, 1\}^*$, for $k \leq n$, acts according to $\Pi_{(\mathbf{h}^C)_k, (\mathbf{h}^\top)_k}$.

²²In the length-preserving proof, the role of Π^{punct} was played by the hash-based argument $\widehat{\Pi}$.

Claim 3.25 (Similarity between f^Π and $f^{\Pi^{\text{punct}}}$).

$$\Pr_{(\mathbf{h}^C, \mathbf{h}^\top, x) \xleftarrow{R} \left(\prod_{j=1}^{p_f(n)} \mathcal{H}_j \right)^2 \times \{0,1\}^n} \left[f^{\Pi_{\mathbf{h}^C, \mathbf{h}^\top}}(x) \neq f^{\Pi^{\text{punct}}_{\mathbf{h}^C, (\mathbf{h}^\top) \leq \text{tr}(n)}}(x) \right] \leq 1/2.$$

for all n with $\ell(n) \geq 2 + \log(p_f^4(p_I(n)))$.

Proof. As in Claim 3.11, we prove the (stronger) statement that the execution traces of $f^{\Pi_{\mathbf{h}^C, \mathbf{h}^\top}}(x)$ and $f^{\Pi^{\text{punct}}_{\mathbf{h}^C, (\mathbf{h}^\top) \leq \text{tr}(n)}}(x)$, for uniform x and $\mathbf{h}^C, \mathbf{h}^\top$, are $1/2$ close. The proof is by coupling the two distributions.

Algorithm 3.26. [Coupling f^Π and $f^{\Pi^{\text{punct}}}$]

1. Sample $(\mathbf{h}^C, \mathbf{h}^\top, x) \xleftarrow{R} \left(\prod_{j=1}^{p_f(n)} \mathcal{H}_j \right)^2 \times \{0,1\}^n$.
2. Let trace and $\text{trace}^{\text{punct}}$ denote the traces of the executions of $f^{\Pi_{\mathbf{h}^C, \mathbf{h}^\top}}(x)$ and $f^{\Pi^{\text{punct}}_{\mathbf{h}^C, (\mathbf{h}^\top) \leq \text{tr}(n)}}(x)$, respectively.
3. Return $(\text{trace}, \text{trace}^{\text{punct}})$.

Let E_i be the event that trace and $\text{trace}^{\text{punct}}$ are identical up to the i^{th} query. We show that for every $i \in [p_f(n)]$ it holds that $\Pr[\neg E_i \mid E_{i-1}] \leq 3\varepsilon(n)$, and since $3\varepsilon(n) \cdot p_f(n) \leq 1/2$, the proof follows by a union bound. Assume E_{i-1} holds. Clearly, the i^{th} query generated by f is the same in both executions. We analyze the probability that the oracle responses differ (i.e., $\neg E_i$ occurs), separately handling prover and verifier queries.

Prover query. Let (ϕ, w) be a query to the prover. By our assumption on f , $(\phi, w) \in \mathcal{R}$. Note that if $|\phi| < \lceil p_f^{-1}(n) \rceil$, then the execution of $\mathbf{P}^{\text{punct}}$ is identical to that of $\mathbf{P}$, and therefore their output is the same. Thus, it remains to handle the case $|\phi| \in [\lceil p_f^{-1}(n) \rceil, p_f(n)]$. On such a query, $\mathbf{P}^{\text{punct}}$ returns $\mathbf{h}^C(\phi)$, whereas the response of $\mathbf{P}$ depends on the success of stInv . Specifically, $\mathbf{P}$ calls $\text{stInv}(1^\lambda, \phi)$ for all $j \in [\lceil p_f^{-1}(|\phi|) \rceil, p_f(|\phi|)]$, and if one of the calls succeeds, it also returns $\mathbf{h}^C(\phi)$. Note that ϕ is a uniform query of f^Π , with respect to uniform choice of $(\mathbf{h}^C, \mathbf{h}^\top, x)$, conditioned on E_{i-1} , and that by the induction assumption, E_{i-1} occurs with probability at least $1/2$. Also recall that by Claim 3.21, the output distribution of qry and the query distribution of f^Π are $1 - \varepsilon(n)$ close, for the considered n (with respect to uniform choice of $(\mathbf{h}^C, \mathbf{h}^\top, x)$). Finally note that, f^Π , via the inversion attempts of Π , calls $\text{stInv}_n(\phi)$. Thus, by Claim 3.19 the inverter stInv finds a preimage and $\mathbf{P}$ returns $\mathbf{h}^C(\phi)$, with probability at least $1 - 2\varepsilon(n) - \varepsilon(n) \geq 1 - 3\varepsilon(n)$.

Verifier query. Let (ϕ, π) be a query to the verifier. If π was generated by a previous prover query, both $\mathbf{V}$ and $\mathbf{V}^{\text{punct}}$ accept. So in the following, we assume π was not generated previously in the execution. If $|\phi| < \lceil p_f^{-1}(n) \rceil$, the execution of $\mathbf{V}^{\text{punct}}$ is *identical* to that of $\mathbf{V}$, and therefore their output is identical. If $|\phi| \geq \lceil p_f^{-1}(n) \rceil$, since $\mathbf{h}^C$ and $\mathbf{h}^\top$ are sampled from a $p_f(p_I(|\phi|))$ -wise independent family for input size $|\phi|$, the probability that $\pi \in \{\mathbf{h}^C(\phi), \mathbf{h}^\top(\phi)\}$ is $2 \cdot 2^{-\ell(|\phi|)} \leq 3\varepsilon(n)$.

Combining the two cases, we deduce that $\Pr[\neg E_i \mid E_{i-1}] \leq 3\varepsilon(n)$. $\square$

3.4.4 The Function g

We are now ready to define the function g , whose security is reduced to the game G . As in the length-preserving proof, we define g to be the weak-to-strong amplification of the function

$g_w: \times_{j=1}^{p_f(n)} \mathcal{H}_j \times \times_{j=1}^{\text{tr}(n)} \mathcal{H}_j \times \{0,1\}^n \rightarrow \times_{j=1}^{p_f(n)} \mathcal{H}_j \times \times_{j=1}^{\text{tr}(n)} \mathcal{H}_j \times \{0,1\}^n$ defined by:

$$g_w(\mathbf{h}^C, \mathbf{h}^T, x) = (\mathbf{h}^C, \mathbf{h}^T, f^{\Pi_{\mathbf{h}^C, \mathbf{h}^T}^{\text{punct}}}(x)) \quad (10)$$

Let g and R_{amp} be the (efficient) function and reduction guaranteed by Lemma 2.6 such that for every PPT A that inverts g with non-negligible probability, $\mathsf{R}_{\text{amp}}^{\mathsf{A}}$ inverts g_w with probability $3/4$.

3.4.5 The Reduction $\hat{R}$

We now define the reduction $\hat{R}$ from the one-wayness of g to the security of G . We use the following wrapper algorithm.

Algorithm 3.27. $[\hat{A}_{\mathbf{h}^C, \mathbf{h}^T}]$

Parameter: $(\mathbf{h}^C, \mathbf{h}^T) \in \{\mathcal{H}_j\}_{j=1}^{p_f(n)} \times \{\mathcal{H}_j\}_{j=1}^{\text{tr}(n)}$.

Oracle: A .

Input: y .

Operation: Return $\mathsf{R}_{\text{amp}}^{\mathsf{A}}(\mathbf{h}^C, \mathbf{h}^T, y)_3$, i.e., the third output of $\mathsf{R}_{\text{amp}}^{\mathsf{A}}$.

Algorithm 3.28. [Reduction $\hat{R}$ from g to G]

Oracle: A .

Input: 1^n .

Operation:

1. Sample $(\mathbf{h}^C, \mathbf{h}^T) \xleftarrow{\mathsf{R}} (\times_{j=1}^{p_f^2(n)} \mathcal{H}_j)^2$.
2. Play in G according to $\mathsf{R}^{\hat{A}_{\mathbf{h}^C, (\mathbf{h}^T)_{\leq n-1}}, \Pi_{\mathbf{h}^C, \mathbf{h}^T}}(1^n)$.

Note that $\hat{R}$ samples the hash functions up to $\mathcal{H}_{p_f^2(n)}$, since the reduction R on input of length n might call the inverter of f with input of length $p_f(n)$. In that inversion, the execution of f might call the interactive argument on inputs of length $p_f(p_f(n))$. The following claim shows that $\hat{R}$ is indeed a reduction from the one-wayness of g to the game G . Let p be the polynomial from Definition 3.1 with respect to R .

Claim 3.29. *For any non-uniform poly-time A that inverts g with probability $1/\text{poly}$ for all $n \in \mathbb{N}$, it holds that $\hat{R}^{\mathsf{A}}$ wins G with probability at least $1/(9p(8))$.*

Proof. The proof follows similar lines to that of Claim 3.15 (used in the length-preserving case), but taking into account that the reduction R might query the inverter of f on arbitrary input lengths.

Let A be a non-uniform poly-time algorithm that inverts g with probability $1/\text{poly}$ for all $n \in \mathbb{N}$. By Lemma 2.6, R_{amp}^A inverts g_w with probability at least $3/4$. Let $(g_w)_y(\mathbf{h}^C, \mathbf{h}^T, x) := g_w(\mathbf{h}^C, \mathbf{h}^T, x)_3$, i.e., it outputs the third output of g_w . By the definition of $\hat{A}$,

$$\Pr_{(\mathbf{h}^C, \mathbf{h}^T, x) \leftarrow (\bigotimes_{j=1}^{p_f(n)} \mathcal{H}_j \times \bigotimes_{j=1}^{\text{tr}(n)} \mathcal{H}_j \times \{0,1\}^n); y \leftarrow (g_w)_y(\mathbf{h}^C, \mathbf{h}^T, x)} \left[(g_w)_y(\mathbf{h}^C, \mathbf{h}^T, \hat{A}_{\mathbf{h}^C, \mathbf{h}^T}(y)) = y \right] \geq 3/4.$$

Hence, by Claim 3.25,

$$\Pr_{(\mathbf{h}^C, \mathbf{h}^T, x) \leftarrow (\bigotimes_{j=1}^{p_f(n)} \mathcal{H}_j)^2 \times \{0,1\}^n; y \leftarrow f^{\Pi_{\mathbf{h}^C, \mathbf{h}^T}}(x)} \left[f^{\Pi_{\mathbf{h}^C, \mathbf{h}^T}}(\hat{A}_{\mathbf{h}^C, (\mathbf{h}^T)_{\leq \text{tr}(n)}}(y)) = y \right] \geq 1/4 \quad (11)$$

for all n with $\ell(n) \geq 2 + \log p_f^4(p_I(n))$. Since we can modify $\hat{A}$ to invert g_w with probability one on all small n , in the following we assume without loss of generality that Equation (11) holds for *all* $n \in \mathbb{N}$. Applying a Markov argument to Equation (11), we get that

$$\Pr_{d_n \leftarrow (\mathbf{h}^C, \mathbf{h}^T) \leftarrow (\bigotimes_{j=1}^{p_f(n)} \mathcal{H}_j)^2} \left[\underbrace{\Pr_{x \leftarrow \{0,1\}^n; y \leftarrow f^{\Pi_{d_n}}(x)} \left[f^{\Pi_{d_n}}(\hat{A}_{\mathbf{h}^C, (\mathbf{h}^T)_{\leq \text{tr}(n)}}(y)) = y \right] \geq 1/8}_{\delta(d_n)} \right] \geq 1/8 \quad (12)$$

Let $(C_G, \mathcal{I})$ be the challenger and the family of sets of the game G , respectively. For $n \in \mathbb{N}$ and $(h^C, h^T) \in \mathcal{H}_n^2$, let C_{h^C, h^T} be the (possibly inefficient) algorithm that on common input 1^n , acts according to V_{h^C, h^T} and outputs 1 if V accepts a statement $x \in \{0,1\}^n \setminus \mathcal{L}_R$ (i.e., the challenger of the game $G_{V, \mathcal{L}_R}$ when defined only for input length n). We claim that for all $k \in \mathbb{N}$ and infinitely many $n \in \mathbb{N}$:

$$\Pr_{d_n \leftarrow (\mathbf{h}^C, \mathbf{h}^T) \leftarrow (\bigotimes_{j=1}^{p_f^2(n)} \mathcal{H}_j)^2; d'_n \leftarrow (\mathbf{h}^C, (\mathbf{h}^T)_{\leq n-1})} \left[\underbrace{\Pr \left[\frac{(\hat{R}^{\hat{A}_{d'_n}, \Pi_{d_n}}, C_{(\mathbf{h}^C)_{n-1}, (\mathbf{h}^T)_{n-1}}(1^n) = 1)}{(\hat{R}^{\hat{A}_{d'_n}, \Pi_{d_n}}, C_G)(1^n, i_n^k) = 1} \geq 1/p(8)}_{\rho(d_n)} \right] \geq 1/8 \right] \quad (13)$$

where the inner probability is over the randomness of R . Indeed, assume Equation (13) does not hold. Hence, by Equation (12),

$$\Pr_{d_n} [\delta(d_n) \wedge \neg \rho(d_n)] > 0$$

for all but finitely many n , where we naturally define $\delta(d_n)$ for $d_n \in (\bigotimes_{j=1}^{p_f^2(n)} \mathcal{H}_j)^2$, by $\delta((d_n)_{\leq p_f(n)})$.

Hence, there exists $\{d_n \leftarrow (\mathbf{h}^C, \mathbf{h}^T) \in (\bigotimes_{j=1}^{p_f^2(n)} \mathcal{H}_j)^2\}_{n \in \mathbb{N}}$ such that $\delta(d_n) = 1$ and $\rho(d_n) = 0$ for all but finitely many n . It follows that there exist a non-uniform interactive argument Π that gets $\{d_n\}_{n \in \mathbb{N}}$ as a non-uniform advice, and a non-uniform algorithm $\hat{A}$ that gets $\{\mathbf{h}^C = (h_1^C, \dots, h_{p_f(n)}^C), \mathbf{h}^T = (h_1^T, \dots, h_{\text{tr}(n)}^T)\}_{n \in \mathbb{N}}$ as a non-uniform advice, such that the following holds:

- $\hat{A}$ inverts f^Π with probability at least $1/8$, for all n .
- $R^{\hat{A}, \Pi}$ does *not* win $G_{V, \mathcal{L}_R}$ nor G , with probability $1/p(8)$.

This, however, contradicts the definition of R , and we conclude that Equation (13) holds.

We next claim that for large enough $n \in \mathbb{N}$ and any $(\mathbf{h}^C, \mathbf{h}_1^\top) \in (\times_{j=1}^{p_f^2(n)} \mathcal{H}_j \times \times_{j=1}^{n-1} \mathcal{H}_j)$:

$$\Pr_{\mathbf{h}_2^\top \leftarrow \times_{j=n}^{p_f^2(n)} \mathcal{H}_j; \mathbf{h}^\top \leftarrow (\mathbf{h}_1^\top, \mathbf{h}_2^\top)} \left[(R^{\hat{A}_{\mathbf{h}^C, \mathbf{h}_1^\top}, \Pi_{\mathbf{h}^C, \mathbf{h}^\top}}, C_{(\mathbf{h}^C)_n, (\mathbf{h}^\top)_n})(1^n) = 1 \right] \leq \text{neg}(n) \quad (14)$$

To see this, first note that since R queries its oracles on input lengths bounded by $p_f(n)$, it invokes $\hat{A}$ on inputs of that length. By construction, on inputs of this length, the function g_w is independent of the function $(\mathbf{h}^\top)_n$. Hence, R gets no information regarding $(\mathbf{h}^\top)_n$ from the oracle calls to $\hat{A}$. Hence, and since R makes fewer than $p_f(n)$ calls to Π , the probability it generates an accepting proof of the form $\mathbf{h}^\top(\phi)$ for a statement ϕ of length n that was not returned by $P_{\mathbf{h}^C, \mathbf{h}^\top}$ is negligible. We also note that, by design, $V_{\mathbf{h}^C, \mathbf{h}^\top}$ does not accept instances $x \notin \mathcal{L}_{\mathcal{R}}$ with a proof of the form $\mathbf{h}^C(x)$ (unless $\mathbf{h}^C(x) = \mathbf{h}^\top(x)$, which occurs with negligible probability).

Combining Equations (13) and (14), we conclude that for all $k \in \mathbb{N}$ and infinitely many $n \in \mathbb{N}$:

$$\Pr_{d_n \leftarrow (\mathbf{h}^C, \mathbf{h}^\top) \leftarrow \times_{j=1}^{p_f^2(n)} \mathcal{H}_j} \left[(R^{\hat{A}_{\mathbf{h}^C, (\mathbf{h}^\top)_{\leq n-1}}, \Pi_{d_n}}, C_G)(1^n, i_n^k) = 1 \right] \geq 1/(8p(8)) - \text{neg}(n).$$

Namely, $\hat{R}^A$ wins G with probability at least $1/(9p(8))$. $\square$

3.4.6 Putting it Together

Proof of Theorem 3.2. The proof follows directly from Claim 3.29. $\square$

3.4.7 Missing proofs

In this section, we prove Lemma 3.20, restated below

Lemma 3.30 (Restatement of Lemma 3.20). *Assume that i.o non-uniform one-way functions do not exist. Then for every $p_E \in \text{poly}$ and $\varepsilon(n) \geq 1/\text{poly}(n)$, there exists $p_I \in \text{poly}$ such that the following holds: for every circuit f_n over $\{0, 1\}^n$ of size $p_E(n)$, there exists a $p_I(n)$ -size circuit C_n such that $\Pr_{x \leftarrow \{0, 1\}^n} [C_n(f_n(x)) \in f_n^{-1}(f_n(x))] \geq 1 - \varepsilon(n)$.*

Proof. Assume i.o non-uniform one-way functions do not exist, and fix $p_E \in \text{poly}$ and $\varepsilon(n) \geq 1/\text{poly}(n)$. We prove that there exists $c \in \mathbb{N}$ such that for every $n \geq c$ and every circuit f_n over $\{0, 1\}^n$ of size $p_E(n)$, there exists an $(n^c + c)$ -size circuit C_n such that $\Pr_{x \leftarrow \{0, 1\}^n} [C_n(f_n(x)) \in f_n^{-1}(f_n(x))] \geq 1 - \varepsilon(n)$. The lemma now follows by considering for every f_n computable in size $p_E(n)$ for input length shorter than c , the circuit of size b that encodes the value of f_n on *all* inputs of that length (b is a constant that depends strictly on c). Therefore, the lemma holds for $p_I = n^c + c + b$.

Assume towards a contradiction that for every $c \in \mathbb{N}$ there exist $n \geq c$ and f_n^c computable in size $p_E(n)$ for input length n , that satisfies the following: for any $(n^c + c)$ -size circuit C_n ,

$$\Pr_{x \leftarrow \{0, 1\}^n} [C_n(f_n^c(x)) \in (f_n^c)^{-1}(f_n^c(x))] < 1 - \varepsilon(n) \quad (15)$$

Let g be the $p_E(n)$ -size circuit family that on input $x \in (\{0,1\}^n)^n$, outputs $(f_n^1(x_1), \dots, f_n^n(x_n))$. Since, by assumption, weak i.o non-uniform one-way functions do not exist, there exist a constant c and an $n^c + c$ -size circuit family $\{C_n\}_{n \in \mathbb{N}}$ such that

$$\Pr_{x \leftarrow (\{0,1\}^n)^n} [C_{n^2}(g_{n^2}(x)) \in g_{n^2}^{-1}(g_{n^2}(x))] > 1 - \varepsilon(n^2) \quad (16)$$

for every $n \in \mathbb{N}$.²³

For $c' \leftarrow 2c$, define the circuit family $C' = \{C'_n\}_{n \geq c'}$ by letting C'_n on input $x \in \{0,1\}^n$, return $(C_{n^2}(x_1, \dots, x_n))_{c'}$, for $x_{c'} \leftarrow x$, where $\{x_j : j \neq c'\}$ are chosen to maximize C_{n^2} inversion probability. It is easy to see that

$$\Pr_{x \leftarrow \{0,1\}^n} [C'_n(f_n^{c'}(x)) \in (f_n^{c'})^{-1}(f_n^{c'}(x))] > 1 - \varepsilon(n^2)$$

for every $n \geq c'$. Since $|C'_n| \leq n^{2c} + c$, this stands in contradiction to the assumption regarding $f_n^{c'}$. $\square$

3.5 Reductions from Auxiliary-Input One-Way Functions

In this section, we explain how to extend our results to reductions with respect to auxiliary-input one-way functions. We start by recalling the notion of (i.o) auxiliary-input one-way functions.

Definition 3.31 (Auxiliary-input one-way functions). *A poly-time function $f: \{0,1\}^{p(n)} \times \{0,1\}^n \rightarrow \{0,1\}^*$ for some $p \in \text{poly}$ is a.i one-way with respect to $\mathcal{I} = \{\mathcal{I}_n \subseteq \{0,1\}^{p(n)}\}_{n \in \mathbb{N}}$, if for any PPT A there exists $\{z_n \in \mathcal{I}_n\}_{n \in \mathbb{N}}$ such that*

$$\Pr_{\substack{x \leftarrow \{0,1\}^n \\ y \leftarrow f_{z_n}(x)}} [f_{z_n}(A(1^n, z_n, y)) = y] \leq \text{neg}(n)$$

for infinitely many n , where $f_{z_n}(\cdot) := f(z_n, \cdot)$. Weak a.i one-way functions and non-uniform a.i one-way functions are analogously defined.

Our main result naturally extends to a.i one-way functions.

Theorem 3.32 (Restatement of Theorem 3.2 with respect to a.i one-way functions). *Let (f, R) be an assumption-dependent fully black-box reduction from an auxiliary-input one-way function with respect to a set $\mathcal{I}$ to non-laconic, non-uniform, ℓ -bit interactive arguments for an NP-relation $\mathcal{R}$, with respect to a game $\mathcal{G}$. Then there exist a non-uniform, poly-time pair of function g and an oracle-aided randomized algorithm $\hat{R}$, and a constant $c > 0$, such that for any non-uniform poly-time Inv that inverts g_{z_n} with probability $1/\text{poly}(n)$ for all $n \in \mathbb{N}$ and $z_n \in \mathcal{I}_n$, algorithm $\hat{R}^{\text{Inv}}$ wins the game $\mathcal{G}$ with probability at least c .*

Moreover, the above holds even if the reduction only needs to work when the interactive argument oracle is perfect zero knowledge.

²³Since the non-uniform function variant of f is not a weak i.o non-uniform one-way function, it is strongly invertible, therefore has a $(1 - \varepsilon(n))$ non-uniform inverter A (which can be converted to an $n^c + c$ -size family of circuits) for any $n \geq n'$ for some $n', c \in \mathbb{N}$. Therefore, as discussed above, it has an $n^{c'} + c'$ -size family of circuits that inverts it with probability $1 - \varepsilon(n)$, for any n , where $c' \geq c$.

The proof of Theorem 3.32 follows very closely to that of Theorem 3.2, while using the following lemma instead of Lemma 3.20.

Lemma 3.33. *Assume that a.i non-uniform one-way functions do not exist. Then for every $p, p_E \in \text{poly}$ and $\varepsilon(n) \geq 1/\text{poly}(n)$, there exists $p_I \in \text{poly}$ such that the following holds: for every circuit $f_n: \{0, 1\}^{p(n)} \times \{0, 1\}^n \rightarrow \{0, 1\}^*$ of size $p_E(n)$ and every $\mathcal{I}_n \subseteq \{0, 1\}^{p(n)}$, there exists a $p_I(n)$ -size circuit C_n such that $\Pr_{x \leftarrow \{0, 1\}^n}^R [C_n(z_n, f_{z_n}(x)) \in f_{z_n}^{-1}(f_{z_n}(x))] \geq 1 - \varepsilon(n)$, for all $z_n \in \mathcal{I}_n$.*

Namely, as in the case of standard one-way functions, if a.i one-way functions do not exist, then any efficient non-uniform function with auxiliary input is strongly invertible.

Given Lemma 3.33, the proof of Theorem 3.32 is a simple adaptation of that of Theorem 3.2.

Proving Lemma 3.33.

Proof of Lemma 3.33. The proof is similar to that of Lemma 3.20, with respect to a.i one-way functions.

Assume a.i non-uniform one-way functions do not exist, and fix $p, p_E \in \text{poly}$ and $\varepsilon(n) \geq 1/\text{poly}(n)$. We prove that there exists $c \in \mathbb{N}$ such that for every $n \geq c$ and every circuit f_n over $\{0, 1\}^{p(n)} \times \{0, 1\}^n$ of size $p_E(n)$ and $\mathcal{I}_n \subseteq \{0, 1\}^{p(n)}$, there exists an $(n^c + c)$ -size circuit C_n such that $\Pr_{x \leftarrow \{0, 1\}^n}^R [C_n(z_n, f_{z_n}(x)) \in f_{z_n}^{-1}(f_{z_n}(x))] \geq 1 - \varepsilon(n)$ for all $z_n \in \mathcal{I}_n$. The lemma now follows by considering for every $\mathcal{I}_n$ and f_n computable in size $p_E(n)$ for input length shorter than c , the circuit of size b that encodes the value of f_n on *all* inputs of that length and all $z_n \in \mathcal{I}_n$ (b is a constant that depends strictly on c). Therefore, the lemma holds for $p_I = n^c + c + b$.

Assume towards a contradiction that for every $c \in \mathbb{N}$ there exist $n \geq c$, set $\mathcal{I}_n^c \subseteq \{0, 1\}^{p(n)}$ and f_n^c computable in size $p_E(n)$ for input length n and auxiliary input length $p(n)$, that satisfies the following: for any $(n^c + c)$ -size circuit C_n , there exists $z_n \in \mathcal{I}_n^c$ such that:

$$\Pr_{x \leftarrow \{0, 1\}^n} [C_n(z_n, f_{z_n}^c(x)) \in (f_{z_n}^c)^{-1}(f_{z_n}^c(x))] < 1 - \varepsilon(n) \quad (17)$$

Let $\mathcal{I} := \{\mathcal{I}_n = \{(z^1, \dots, z^n) \in \mathcal{I}_n^1 \times \dots \times \mathcal{I}_n^n\}\}_{n \in \mathbb{N}}$ and let g be the $p_E(n)$ -size circuit family that on input $x \in (\{0, 1\}^n)^n$ and auxiliary input $z_n = (z^1, \dots, z^n) \in \mathcal{I}_n^{24}$, outputs $(f_{z^1}^1(x_1), \dots, f_{z^n}^n(x_n))$. Since, by assumption, weak a.i non-uniform one-way functions do not exist, there exist a constant c and an $n^c + c$ -size circuit family $\{C_n\}_{n \in \mathbb{N}}$ such that

$$\Pr_{x \leftarrow (\{0, 1\}^n)^n} [C_{n^2}(z_n, g_{z_n}(x)) \in g_{z_n}^{-1}(g_{z_n}(x))] > 1 - \varepsilon(n^2) \quad (18)$$

for every $n \in \mathbb{N}$ and every $z_n \in \mathcal{I}_n$.²⁵

For $c' \leftarrow 2c$, define the circuit family $C' = \{C'_n\}_{n \geq c'}$ by letting C'_n on input $x \in \{0, 1\}^n$ and auxiliary input $z_n \in \mathcal{I}_n^{c'}$, return $(C_{n^2}((z_n^1, \dots, z_n^n), (x_1, \dots, x_n)))_{c'}$, for $x_{c'} \leftarrow x$ and $z_{n'}^{c'} \leftarrow z_n$,

²⁴On input length n^2 , the function g expects to receive auxiliary input of size $p(n^2)$ and not $n \cdot p(n)$, as defined. To handle it, assume without loss of generality that $p(n^2) \geq n \cdot p(n)$, and that g receives extra auxiliary information which it ignores.

²⁵Since the non-uniform function variant of f is not weak a.i one-way, it has a $(1 - \varepsilon(n))$ non-uniform inverter A (which can be converted to an $n^c + c$ -size family of circuits) for any $n \geq n'$ and any $z_n \in \mathcal{I}_n$, for some $n', c \in \mathbb{N}$. Therefore, it has a $n^{c'} + c'$ -size family of circuits that inverts it for *any* n and any $z_n \in \mathcal{I}_n$.

where $\{x_j: j \neq c'\}$ are chosen to maximize C_{n^2} 's inversion probability, and $\{z_n^j: j \neq c'\}$ are fixed to arbitrary values. It is easy to see that

$$\Pr_{x \leftarrow \{0,1\}^n} \left[C'_n(z_n, f_{z_n}^{c'}(x)) \in (f_{z_n}^{c'})^{-1}(f_{z_n}^{c'}(x)) \right] > 1 - \varepsilon(n^2)$$

for every $n \geq c'$ and $z_n \in \mathcal{I}_n^{c'}$. Since $|C'_n| \leq n^{2c} + c$, this contradicts the assumption regarding $f_n^{c'}$. $\square$

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